# Covert Capacity-Key Tradeoff over Discrete Memoryless Networks

Abdelaziz Bounhar, Mireille Sarkiss, Michèle Wigger

***Abstract*—This paper explores the tradeoff between covert communication rates and secret key usage in discrete memoryless channels (DMCs), multiple-access channels (DMMACs), and interference channels (DMICs). In particular, it determines the highest covert rates that are achievable with a given secret key budget. This forms what we call the covert capacity-key tradeoff.**

**Our results show that when secret key resources are limited and the adversary has a stronger channel than the legitimate receiver(s), simple binary signaling is optimal at all transmitters, regardless of the channel or input alphabets. In this regime, the covert capacity grows linearly with the key rates. Moreover, for DMMACs and DMICs, each user can independently reach their own maximal covert rate without affecting others, leading to a square-shaped tradeoff region. At higher key rates or when the adversary is weaker for certain inputs, a more complex interaction emerges between users rates. Moreover, binary signaling is no longer optimal and the covert capacity-key tradeoff region grows sublinearly.**

***Index Terms*—Covert communication, capacity-key tradeoff, discrete memoryless channel, multiple access channels, interference channels.**

## I. Introduction

As cyber threats continue to evolve, research into advanced security mechanisms remains essential particularly for securing communication between distant parties. One important area of secure communication is physical layer security, which differs from traditional cryptographic approaches that rely on computational complexity to protect information. Instead, physical layer security leverages the inherent properties of wireless channels to prevent adversaries from intercepting or decoding transmissions.

Covert communication aims to hide not just the content of a message, but also the fact that communication is happening at all. This is critical in settings like cybersecurity or military operations, where revealing the act of communication alone can be risky. Unlike traditional encryption, covert communication blends signals into background noise, often using sparse transmissions, which renders detection difficult. Sparse transmissions moreover have little power consumption, making covert communication an attractive candidate to establish security in the IoT.

The main limitation of covert communication is its capacity: the number of bits that can be sent undetectably scales only with the square root of the blocklength—a result known as the square-root law. This fundamental limit has been established across many channel models [1]–[4], and is only violated for some degenerated channels, e.g., when the adversary has uncertainty [5]–[8] or when cooperative jamming is used [9]–[11]. More recent work has explored covert communication in multi-user settings [12]–[15], such as over broadcast, multiple-access, and interference channels.

A common theme across these results is that achieving positive covert rates often requires transmitters and legitimate receivers to share a secret key unknown to the adversary. However, in practical systems, generating and distributing such keys incurs an overhead, so it is desirable to minimize the required key length. Previous research, including [2], [13], [14], determined the minimum key rates (measured as the number of key bits divided by the square-root of the blocklength and the covertness constraint) required to achieve the largest possible covert rates. However, a crucial question remained unanswered:

- What is the minimum key rate required to achieve a given covert rate?
- Which covert rates are achievable given a limited key budget?

This quantity was first studied in [15] for discrete memoryless channels (DMCs) with binary inputs, where it was termed the *covert capacity-key tradeoff*. It was shown in [15] that for binary-input DMCs, this covert capacity-key tradeoff follows a linear growth for small key rates before saturating at the *covert capacity*, i.e., the largest achievable covert rate under an unlimited key budget.

In this paper, we fully characterize this covert capacity-key tradeoff for DMCs with arbitrary input alphabets. We show that when the adversary has a uniformly stronger channel, the tradeoff grows linearly at low key rates, and binary signaling is optimal—even for non-binary alphabets. In contrast, if the adversary is weaker or varies across inputs, the tradeoff becomes sublinear and more complex signaling strategies are required.

We extend these results to multi-user channels, including two-user DMMACs and DMICs. We provide the first full description of the covert capacity-key tradeoff region in these settings. Notably, we find that when the adversary is strong and key rates are low, each user can independently achieve their highest covert rate, meaning there's no tradeoff and binary signaling is optimal at all transmitters. But as key rates increase or the adversary becomes weaker than the legitimate receiver for some inputs, the users' maximum rates begin to interact.

*Notation*: In this paper, we follow standard information theory notations. We use calligraphic fonts for sets (e.g. $\mathcal{S}$) and note by $|\mathcal{S}|$ the cardinality of a set $\mathcal{S}$. Random variables are denoted by upper case letters (e.g., $X$), while their realizations are denoted by lowercase letters (e.g. $x$). We write $X^n$ and $x^n$ for the tuples $(X_1, \ldots, X_n)$ and $(x_1, \ldots, x_n)$, respectively, for

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any positive integer $n > 0$. For a distribution $P$ on $\mathcal{X}$, we note its product distribution on $\mathcal{X}^n$ by $P^{\otimes n}(x^n) = \prod_{i=1}^n P(x_i)$. We also denoteby $\text{Supp}(P)$ the support of a distribution $P$, i.e. $\text{Supp}(P) = \{x \colon P(x) \neq 0\}$. For two distributions $P$ and $Q$ on $\mathcal{X}$, $\mathbb{D}(P\|Q) = \sum_{x\in\mathcal{X}} P(x) \log\left(\frac{P(x)}{Q(x)}\right)$ is the relative entropy. The logarithm function is in base 2 and motivated by continuity of the function $t \log t$ we define $0 \log(0) = 0$. We use Landau notation, i.e., for a function $f(n)$ we write $f(n) = o(g(n))$ if the ratio $f(n)/g(n)$ vanishes as $n \to \infty$, and we write $f(n) = \mathcal{O}(g(n))$ if the cumulation points of the ratio $f(n)/g(n)$ are within a bounded interval.

We abbreviate *probability mass function* by *pmf* and *independent and identically distributed* by *i.i.d.*.

## II. The Single-User Setup

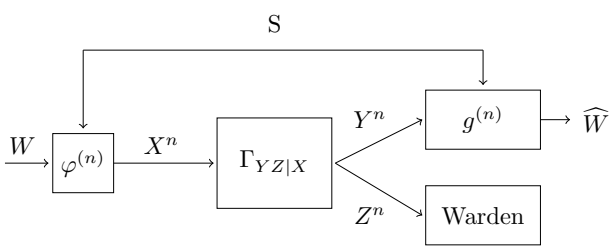


Fig. 1: Point-to-point covert communication over a Discrete Memoryless Channel of law $\Gamma_{YZ|X}$.

Consider the setup illustrated in Figure 1. The transmitter (Tx) wishes to send a message $W$ to the legitimate receiver (Rx) while avoiding detection by the warden which attempts to detect the presence of communication. Communication takes place over a block of $n$ channel uses. The Tx produces channel inputs in a finite alphabet $\mathcal{X}$ and the legitimate Rx and the warden observe channel outputs within finite alphabets $\mathcal{Y}$ and $\mathcal{Z}$. These outputs are produced by a DMC, that means, if the Tx produces the $n$ channel inputs $X^n = x^n$ then for any $i \in \{1, \dots, n\}$ the $i$-th output symbols $Y_i$ and $Z_i$ observed at the legitimate Rx and the warden are generated from the $i$-th input $x_i$ according to the conditional laws $\Gamma_{Y|X}(\cdot|x_i)$ and $\Gamma_{Z|X}(\cdot|x_i)$, repectively.

The Tx encodes message $W$ using some encoding function $\varphi^{(n)}$ defined on appropriate domains, along with a secret-key $S$. Subsequently, it sends the resulting codeword

$$X^n = \varphi^{(n)}(W, S) \tag{1}$$

over the channel. For readability, we will write $x^n(w, s)$ instead of $\varphi^{(n)}(w, s)$. Let the message $W$ and the secret-key $S$ be represented by two sequences of $m$ and $p$ i.i.d. Bernoulli-$1/2$ bits, where these numbers $m$ and $p$ will depend on the blocklength $n$. The secret-key $S$ is exclusively known to the transmitter and the Rx but not to the warden.

The legitimate Rx estimates the message as:

$$\widehat{W} = g^{(n)}(Y^n, S) \tag{2}$$

using an appropriate decoding function.

To ensure reliability of communication, we seek for systems (encoding and decoding functions) where

$$\lim_{n\to\infty} \Pr[\widehat{W} \neq W] = 0. \tag{3}$$

At the same time we impose that the output distribution implied at the warden

$$\widehat{Q}^n(z^n) \triangleq \frac{1}{2^m 2^p} \sum_{(w,s)} \Gamma_{Z|X}^{\otimes n}(z^n|x^n(w, s)). \tag{4}$$

be almost indistinguishable from the warden's output distribution when the all-zero sequence is transmitted (which stands for absence of communication), i.e., from

$$\Gamma_{Z|X}^{\otimes n}(z^n|0^n). \tag{5}$$

Notice that we impose that the 0-symbol be part of the input alphabet $\mathcal{X}$ and then require that the Kullback-Leibler (KL) divergence

$$\delta_n \triangleq \mathbb{D}\left(\widehat{Q}^n(\cdot)\middle\|\Gamma_{Z|X}^{\otimes n}(\cdot|0^n)\right) \tag{6}$$

vanishes in the regime of large blocklengths $n \to \infty$.

Our goal is to determine the largest data rate in function of the available key rate. In covert-communication tradition [1]–[3] rates are obtained by scaling the number of message or secret key bits with the square-root of the number of channel uses and the divergence-$\delta_n$. This yields the following definition.

*Definition 1:* For any given $k \geq 0$, define the *covert capacity-key tradeoff* $r^\star(k)$ as the largest rate $r$ such that there exists a sequence (in the blocklength $n$) of tuples $(m, p)$ and encoding/decoding functions $(\varphi^{(n)}, g^{(n)})$ satisfying

$$\lim_{n\to\infty} \Pr[\widehat{W} \neq W] = 0, \tag{7a}$$

$$\lim_{n\to\infty} \delta_n = 0, \tag{7b}$$

and

$$r \leq \liminf_{n\to\infty} \frac{m}{\sqrt{n\delta_n}}, \tag{8a}$$

$$k \geq \limsup_{n\to\infty} \frac{p}{\sqrt{n\delta_n}}. \tag{8b}$$

To ensure that the problem is non-degenerate, we assume:

$$\sum_{x\in\mathcal{X}\setminus\{0\}} \psi(x)\Gamma_{Z|X}(\cdot|x) \neq \Gamma_{Z|X}(\cdot|0), \quad \forall \psi(\cdot), \tag{9a}$$

$$\text{Supp}\left(\Gamma_{Z|X}(\cdot|x)\right) \subseteq \text{Supp}\left(\Gamma_{Z|X}(\cdot|0)\right), \quad \forall x \in \mathcal{X}, \tag{9b}$$

$$\text{Supp}\left(\Gamma_{Y|X}(\cdot|x)\right) \subseteq \text{Supp}\left(\Gamma_{Y|X}(\cdot|0)\right), \quad \forall x \in \mathcal{X} \tag{9c}$$

where in the above, $\psi(\cdot)$ indicates any pmf over $\mathcal{X}\setminus\{0\}$.

Under these assumptions, we define the covert capacity as:

$$C_{\text{covert}} := \sup_{k\geq 0} r^\star(k). \tag{10}$$

Notice that if (9a) is violated, the transmitter can trivially send information at positive traditional rates (i.e., the number of message bits scaling with the blocklength $n$) without being detected. In other words, if (9a) is violated, the covert capacity is trivially $r^\star(k) = \infty$ for all positive key rates $k > 0$. If (9b) is violated, then with high probability the warden can detect communication from the Tx whenever certain symbols are employed. In this case, it is clear that the Tx should not use any of the inputs violating (9b), reducing the problem to a setup where (9b) holds. If (9c) is violated, then the legitimate Rx gets can identify certain inputs where non-zero symbols have been transmitted. In this case, the covert capacity is again $\infty$.

### A. Results

Given a pmf $\psi(\cdot)$ over $\mathcal{X}\backslash\{0\}$, we use the abbreviations

$$\chi^2(\psi) \triangleq \sum_{z\in\mathcal{Z}} \frac{\left(\sum_{x\in\mathcal{X}} \psi(x)\Gamma_{Z|X}(z|x) - \Gamma_{Z|X}(z|0)\right)^2}{\Gamma_{Z|X}(z|0)}. \quad (11)$$

Moreover, for given $x \in \mathcal{X}\backslash\{0\}$, we define:

$$\mathbb{D}_Y(x) \triangleq \mathbb{D}\left(\Gamma_{Y|X}(\cdot|x)\|\Gamma_{Y|X}(\cdot|0)\right), \quad (12)$$

$$\mathbb{D}_Z(x) \triangleq \mathbb{D}\left(\Gamma_{Z|X}(\cdot|x)\|\Gamma_{Z|X}(\cdot|0)\right). \quad (13)$$

Finally, we define a function $k \mapsto f(\psi, k)$ for each choice of the pmf $\psi(\cdot)$. The definition of $k \mapsto f(\psi, k)$ depends on whether the difference

$$\Delta_\psi \triangleq \sum_{x\in\mathcal{X}\backslash\{0\}} \psi(x)\cdot(\mathbb{D}_Z(x) - \mathbb{D}_Y(x)) \quad (14)$$

is positive or not. For pmfs $\psi$ for which $\Delta_\psi > 0$, we define

$$f(\psi,k) = \min\left\{\frac{\sum_{x\in\mathcal{X}\backslash\{0\}}\psi(x)\cdot\mathbb{D}_Y(x)}{\sum_{x\in\mathcal{X}\backslash\{0\}}\psi(x)\cdot(\mathbb{D}_Z(x)-\mathbb{D}_Y(x))}\cdot k, \sqrt{2}\frac{\sum_{x\in\mathcal{X}\backslash\{0\}}\psi(x)\cdot\mathbb{D}_Y(x)}{\sqrt{\chi^2(\psi)}}\right\}. \quad (15)$$

and for pmfs $\psi$ for which $\Delta_\psi \leq 0$, we define

$$f(\psi,k) = C_\psi \triangleq \sqrt{2}\frac{\sum_{x\in\mathcal{X}\backslash\{0\}}\psi(x)\cdot\mathbb{D}_Y(x)}{\sqrt{\chi^2(\psi)}}, \quad (16)$$

So, if $\Delta_\psi \leq 0$ the function is constant equal to $C_\psi$. If $\Delta_\psi > 0$, the function is a straight line from the origin $(k = 0, r = 0)$ to the point

$$k = k_\psi \triangleq \sqrt{2}\frac{\sum_{x\in\mathcal{X}\backslash\{0\}}\psi(x)\cdot(\mathbb{D}_Z(x)-\mathbb{D}_Y(x))}{\sqrt{\chi^2(\psi)}} \quad (17)$$

$$r = C_\psi \triangleq \sqrt{2}\frac{\sum_{x\in\mathcal{X}\backslash\{0\}}\psi(x)\cdot\mathbb{D}_Y(x)}{\sqrt{\chi^2(\psi)}}, \quad (18)$$

and is constant equal to $f(\psi, k) = C_\psi$ for all key rates $k \geq k_\psi$.

We can now state the covert capacity-key tradeoff of a DMC:

*Theorem 1:* We have

$$r^\star(k) = \max_\psi f(\psi, k), \quad (19)$$

where the maximum is over all pmfs $\psi(\cdot)$ on $\mathcal{X}\backslash\{0\}$. Moreover,

$$C_{\text{covert}} := \sup_\psi C_\psi. \quad (20)$$

*Proof:* Follows as a special case from Theorem 5 ahead. ■

For the further analysis of $r^\star(k)$, we distinguish three cases depending on whether $\Delta_\psi$ is positive for all pmfs $\psi$, is negative for all $\psi$, or it is positive for some $\psi$s and negative for others.

Consider an example where $\Delta_\psi$ is always positive:

*Example 1 (Adversary Stronger than Legitimate Receiver):* Consider a channel with input alphabet $\mathcal{X} = \{0, 1, 2\}$, output alphabets $\mathcal{Z} = \mathcal{Y} = \{0, 1, 2, 3, 4\}$, and transition pmfs:

$$\Gamma_{Y|X} = \begin{bmatrix} 0.23, & 0.20, & 0.25, & 0.05, & 0.27 \\ 0.35, & 0.22, & 0.10, & 0.05, & 0.28 \\ 0.27, & 0.24, & 0.24, & 0.07, & 0.18 \end{bmatrix} \quad (21a)$$

and

$$\Gamma_{Z|X} = \begin{bmatrix} 0.22, & 0.32, & 0.15, & 0.12, & 0.19 \\ 0.36, & 0.03, & 0.39, & 0.21, & 0.01 \\ 0.31, & 0.20, & 0.07, & 0.22, & 0.20 \end{bmatrix} \quad (21b)$$

where the rows pertain to the three input symbols $x = 0, 1, 2$ and the five columns to the five output symbols $0, 1, 2, 3, 4$. For

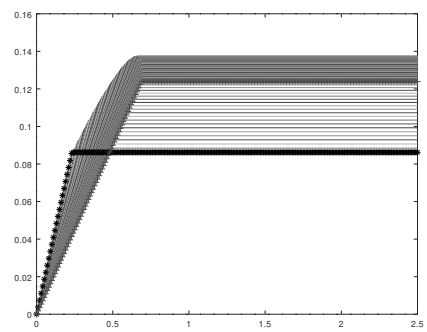


Fig. 2: Functions $k \mapsto f(\psi, k)$ for different pmfs $\psi$.

this channel the difference $\Delta_\psi > 0$ for all pmfs $\psi$, since for all $x \in \mathcal{X}\backslash\{0\}$ we have $\mathbb{D}_Z(x) - \mathbb{D}_Y(x) > 0$. Figure 2 illustrates the function $k \mapsto f(\psi, k)$ for different choices of the input pmf $\psi(\cdot)$. The line with $+$ markers corresponds to the degenerate choice $\psi(1) = 1$ and the line with $*$-markers to the degenerate choice $\psi(2) = 1$.

We observe that in this example the extreme lines $f(\psi, k)$ with smallest and largest slopes correspond to the degenerate pmfs that put probability 1 on a single non-zero input. This is true in general as can be seen by the following lemma and noting that the slope of $f(\psi, k)$ is given by

$$S_\psi \triangleq \frac{\sum_{x\in\mathcal{X}\backslash\{0\}}\psi(x)\cdot\mathbb{D}_Y(x)}{\sum_{x\in\mathcal{X}\backslash\{0\}}\psi(x)\cdot(\mathbb{D}_Z(x)-\mathbb{D}_Y(x))} > 0. \quad (22)$$

*Lemma 1:* If

$$\mathbb{D}_Z(x) - \mathbb{D}_Y(x) > 0, \qquad \forall x \in \mathcal{X}\backslash\{0\}, \quad (23)$$

then $S_\psi > 0$ for all input pmfs $\psi(\cdot)$ and is largest (smallest) for a degenerate pmf $\psi(\cdot)$ that puts probability mass 1 on one of the non-zero inputs.

*Proof:* Consider the convex hull of the points

$$\{(\mathbb{D}_Z(x) - \mathbb{D}_Y(x),\ \mathbb{D}_Y(x))\}_{x\in\mathcal{X}\backslash\{0\}} \quad (24)$$

which by our assumption all lie in the first quadrant of the two-dimensional Euclidean space. If for each point we consider the straight line from the origin to this point, the line with largest (smallest) slope is attained at one of the points in (24). Moreover, the slope for each of the points in (24) is given by

$$\frac{\mathbb{D}_Y(x)}{\mathbb{D}_Z(x) - \mathbb{D}_Y(x)}, \quad (25)$$

and thus equal to $S_\psi$ for degenerate pmf $\psi$ with probability mass 1 on the non-zero symbol $x$.

In a similar way, since each point $P = (P_1, P_2)$ in the convex hull is naturally assigned to a different pmf $\psi$ in the sense that:

$$P_1 = \sum_{x\in\mathcal{X}\backslash\{0\}} \psi(x)\cdot(\mathbb{D}_Z(x) - \mathbb{D}_Y(x)) \quad (26)$$

$$P_2 = \sum_{x \in \mathcal{X}\backslash\{0\}} \psi(x) \cdot \mathbb{D}_Y(x), \tag{27}$$

and the slope of the straight line from the origin to this $P$ is given by $S_\psi$, from the arguments in the previous paragraphs, we can deduce that the largest (smallest) slope $S_\psi$ is obtained from a degenerate pmf $\psi$ with mass on a single non-zero symbol. ■

With above lemma, we can simplify Theorem 1 for the class of channels satisfying (23).

*Corollary 2 (Adversary Stronger than Legitimate Receiver):* If (23) holds, then

$$r^\star(k) = \begin{cases} S_{\max} \cdot k & \text{if } \in [0,\ k_{\text{lin}}]\,, \\ \max_{\psi \in \mathcal{L}(k)} C_\psi & \text{if } \in (k_{\text{lin}},\ k_{\text{sat}})\,, \\ C_{\text{covert}} & \text{if } \in [k_{\text{sat}}, \infty)\,, \end{cases} \tag{28}$$

where

$$S_{\max} \triangleq \max_{x \in \mathcal{X}\backslash\{0\}} \frac{\mathbb{D}_Y(x)}{\mathbb{D}_Z(x) - \mathbb{D}_Y(x)}; \tag{29}$$

and

$$k_{\text{lin}} \triangleq \sqrt{2}\frac{\mathbb{D}_Z(x_{\text{best}}) - \mathbb{D}_Y(x_{\text{best}})}{\sqrt{\chi^2(\delta_{x_{\text{best}}})}} \tag{30}$$

$$k_{\text{sat}} \triangleq \sqrt{2}\frac{\sum_{x \in \mathcal{X}\backslash\{0\}} \psi^*(x) \cdot \mathbb{D}_Y(x)}{\sqrt{\chi^2(\psi^*)}}, \tag{31}$$

where $x_{\text{best}}$ is the maximizer in (29); $\delta_{x_{\text{best}}}$ indicates the degenerate pmf with probability 1 at $x_{\text{best}}$; and $\psi^*$ the maximizer in (20), i.e., the covert capacity achieving input-distribution. Moreover,

$$\mathcal{L}(k) \triangleq \left\{ \psi\colon \quad k \geq \sqrt{2}\frac{\sum_{x \in \mathcal{X}\backslash\{0\}} \psi(x) \cdot \mathbb{D}_Y(x)}{\sqrt{\chi^2(\psi)}} \right\}. \tag{32}$$

*Remark 1 (Optimality of Binary Signaling):* From Lemma 1 and above Corollary 2, we can directly deduce that for channels satisfying (23), binary signaling on inputs 0 and $x_{\text{best}}$ is optimal for small key rates $k \leq S_{\max}$, where recall that $x_{\text{best}}$ is the maximizing input in (29) and $S_{\max}$ is defined in (29). Moreover, in the regime of small key rates the covert capacity-key tradeoff is linear. Specifically, increasing the key-rate by 1 will increase the largest achievable rate by $S_{\max}$.

For channels where (23) holds, the capacity-key tradeoff starts at the origin: $r^\star(0) = 0$. For all other channels however this is not the case.

In fact, when the adversary is no better than the legitimate Rx for at least one of the non-zero inputs, i.e., when

$$\mathbb{D}_Z(x) - \mathbb{D}_Y(x) \leq 0, \qquad \text{for some } x \in \mathcal{X}\backslash\{0\}, \tag{33}$$

then a positive message-rate is achievable even with zero key-rate, i.e., $r^\star(0) > 0$. We have the following corollary directly from Theorem 1 and our discussion above.

*Corollary 3 (Adversary Sometimes Weaker than Legitimate Rx):* If (33) holds, then

$$r^\star(k) = \max_{\psi \in \mathcal{L}(k)} \sqrt{2}\frac{\sum_{x \in \mathcal{X}\backslash\{0\}} \psi(x) \cdot \mathbb{D}_Y(x)}{\sqrt{\chi^2(\psi)}}. \tag{34}$$

In particular,

$$r^\star(k) = C_{\text{covert}}, \qquad k \geq k_{\text{sat}}. \tag{35}$$

The following example illustrates the covert capacity tradeoff for a channel not satisfying Assumption (23).

*Example 2 (Adversary Sometimes Weaker than Legitimate Rx):* Consider a channel with ternary inputs $\mathcal{X} = \{0, 1, 2\}$,

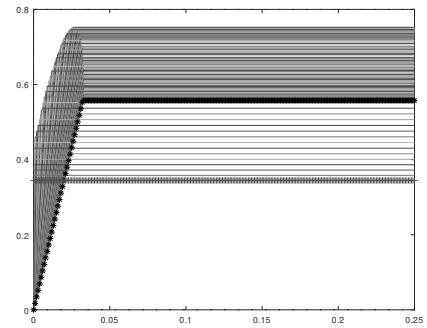


Fig. 3: Functions $k \mapsto f(\psi, k)$ for different pmfs $\psi$.

quinary outputs $\mathcal{Y} = \mathcal{Z} = \{0, 1, 2, 3, 4\}$, and transition pmfs:

$$\Gamma_{Y|X} = \begin{bmatrix} 0.24 & 0.10 & 0.22 & 0.22 & 0.22 \\ 0.20 & 0.14 & 0.26 & 0.328 & 0.072 \\ 0.06 & 0.19 & 0.2 & 0.05 & 0.50 \end{bmatrix} \tag{36a}$$

and

$$\Gamma_{Z|X} = \begin{bmatrix} 0.32 & 0.22 & 0.23 & 0.13 & 0.10 \\ 0.47 & 0.25 & 0.10 & 0.14 & 0.04 \\ 0.38 & 0.01 & 0.15 & 0.12 & 0.34 \end{bmatrix} \tag{36b}$$

For this channel, we have $\mathbb{D}_Z(1) - \mathbb{D}_Y(1) < 0$ but $\mathbb{D}_Z(2) - \mathbb{D}_Y(2) > 0$. Figure 3 illustrates the function $k \mapsto f(\psi, k)$ for above channel and for different choices of the input pmf $\psi(\cdot)$. The convex hull of the lines indicates the covert capacity-key tradeoff $r^\star(k)$.

In the last case where

$$\mathbb{D}_Z(x) - \mathbb{D}_Y(x) \leq 0, \qquad \text{for all } x \in \mathcal{X}\backslash\{0\}, \tag{37}$$

we also have for any $\psi$:

$$\sum_{x \in \mathcal{X}\backslash\{0\}} \psi(x)\left(\mathbb{D}_Z(x) - \mathbb{D}_Y(x)\right) \leq 0 \tag{38}$$

and the functions $k \mapsto f(\psi, k)$ are straigth lines. Accordingly, $r^\star(k)$ is constant: $r^\star(k) = C_{\text{covert}}$ for all $k \geq 0$. we obtain the following corollary.

*Corollary 4 (Adversary Weaker than Legitimate Rx):* Under Condition (37),

$$r^\star(k) = C_{\text{covert}}, \quad k \geq 0. \tag{39}$$

*Remark 2 (Special Case of Binary Inputs):* for binary inputs there is only a single choice of $\psi$, namely putting all mass on the single non-zero symbol. In this case:

- If $\mathbb{D}_Z(x) - \mathbb{D}_Y(x) \leq 0$ for $x$ the single non-zero symbol, then $r^\star(k)$ is constant:

$$r^\star(k) = C_{\text{covert}}, \quad k \geq 0. \tag{40}$$

- Else, $r^\star(k)$ is linearly increasing in $k$ until it saturates at $C_{\text{covert}}$:

$$r^\star(k) = \min\left\{ \frac{\mathbb{D}_Y(x)}{\mathbb{D}_Z(x) - \mathbb{D}_Y(x)} \cdot k,\ C_{\text{covert}} \right\}. \tag{41}$$

## III. The Multiple-Access Channel

We turn our focus to a two-user multi-access channel (MAC), see Figure 4, where the two Txs produce inputs in finite alphabets $\mathcal{X}_1$ and $\mathcal{X}_2$, the legitimate Rx observes outputs in the finite alphabet $\mathcal{Y}$, and the warden observes outputs in the finite alphabet $\mathcal{Z}$.

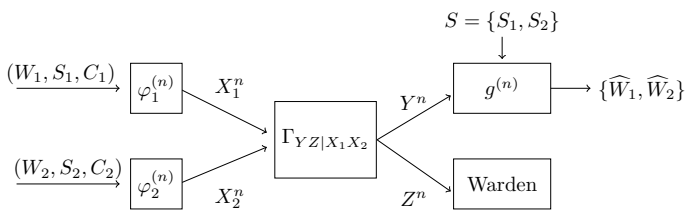


Fig. 4: Covert communication over a two-user MAC.

This covert MAC has previously been studied in [13]. The main difference here is that each Tx $j \in \{1, 2\}$ has access to additional *local* randomness described by $C_i$ which consists of $g_j$ i.i.d. Bernoulli-$1/2$ bits. Each Tx $j \in \{1, 2\}$ thus produces its channel inputs as

$$X_j^n = \varphi_j^{(n)}(W_j, S_j, C_j), \tag{42}$$

where $W_j$ and $S_j$ are independent i.i.d. Bernoulli-$1/2$ bitstrings of lengths $m_j$ and $p_j$ (which grow with $n$), respectively and $S_j$ is know to Tx $j$ and the Rx, while $W_j$ and $C_j$ only to Tx $j$.

As for the single-user setup, the legitimate Rx and the warden observe outputs generated by discrete memoryless channels $\Gamma_{Y|X_1X_2}(\cdot|\cdot,\cdot)$ and $\Gamma_{Z|X_1X_2}(\cdot|\cdot,\cdot)$ based on the input sequences produced at the two Txs. That means, if Tx 1 sends inputs $X_1^n = x_1^n$ and Tx 2 sends $X_2^n = x_2^n$, then for each $i \in \{1, \ldots, n\}$, the legitimate Rx observes output symbol $Y_i$ following the conditional law $\Gamma_{Y|X_1X_2}(\cdot|x_{1,i}, x_{2,i})$ and the adversary observes output $Z_i$ following the conditional law $\Gamma_{Z|X_1X_2}(\cdot|x_{1,i}, x_{2,i})$.

After observing outputs $Y^n$, the legitimate Rx decodes both messages $W_1$ and $W_2$ as:

$$(\widehat{W}_1, \widehat{W}_2) = g^{(n)}(Y^n, S_1, S_2). \tag{43}$$

using an appropriate decoding function $g^{(n)}$.

Covertness imposes that the output distribution at the warden

$$\widehat{Q}^n(z^n) \triangleq \frac{\sum\limits_{\substack{(w_1,w_2,s_1,\\ s_2,c_1,c_2)}} \Gamma_{Z|X_1X_2}^{\otimes n}(z^n|x_1^n(w_1,s_1,c_1), x_2^n(w_2,s_2,c_2))}{2^{m_1+m_2}2^{p_1+p_2}2^{g_1+g_2}}. \tag{44}$$

be almost indistinguishable from the distribution when the Txs send only the zero-symbol, which is assumed to belong to both input alphabets $\mathcal{X}_1$ and $\mathcal{X}_2$. Technically speaking, covertness requires that the KL-divergence

$$\delta_n \triangleq \mathbb{D}\left(\widehat{Q}^n \middle\| \Gamma_{Z|X_1X_2}^{\otimes n}(\cdot|0^n, 0^n)\right) \tag{45}$$

vanishes in the regime of large blocklengths.

*Definition 2:* For given $k_1, k_2 \geq 0$, define the *capacity-key tradeoff region* $\mathcal{R}^\star(k_1, k_2)$ as the set of all rate pairs $(r_1, r_2)$ such there exists a sequence (in the blocklength $n$) of tuples $(m_1, m_2, p_1, p_2, g_1, g_2)$ and encoding/decoding functions $(\varphi_1^{(n)}, \varphi_2^{(n)}, g^{(n)})$ satisfying

$$\lim_{n\to\infty} \Pr[(\widehat{W}_1, \widehat{W}_2) \neq (W_1, W_2)] = 0, \tag{46a}$$

$$\lim_{n\to\infty} \delta_n = 0, \tag{46b}$$

and

$$r_j \leq \liminf_{n\to\infty} \frac{m_j}{\sqrt{n\delta_n}}, \quad j \in \{1, 2\}, \tag{47a}$$

$$k_j \geq \limsup_{n\to\infty} \frac{p_j}{\sqrt{n\delta_n}}, \quad j \in \{1, 2\}. \tag{47b}$$

Define further

$$\tilde{\mathcal{X}} \triangleq ((\mathcal{X}_1 \backslash \{0\}) \times \{0\}) \cup (\{0\} \times (\mathcal{X}_1 \backslash \{0\})). \tag{48}$$

To avoid that the problem be trivial or impossible, we impose the following restrictions on the MAC to hold for any input pair $(x_1, x_2) \in \tilde{\mathcal{X}}$:

$$\sum_{\psi} \psi(x_1, x_2) \Gamma_{Z|X_1X_2}(\cdot|x_1, x_2) \neq \Gamma_{Z|X_1X_2}(\cdot|0, 0), \quad \forall \psi, \tag{49a}$$

$$\text{Supp}(\Gamma_{Y|X_1X_2}(\cdot|x_1, x_2)) \subseteq \text{Supp}(\Gamma_{Y|X_1X_2}(\cdot|0, 0)) \quad \forall x_1 \in \mathcal{X}, x_2 \in \mathcal{X}_2 \tag{49b}$$

$$\text{Supp}(\Gamma_{Z|X_1X_2}(\cdot|x_1, x_2)) \subseteq \text{Supp}(\Gamma_{Z|X_1X_2}(\cdot|0, 0)), \quad \forall x_1 \in \mathcal{X}, x_2 \in \mathcal{X}_2. \tag{49c}$$

where here $\psi$ denotes any pmf over $(\mathcal{X}_1 \times \mathcal{X}_2) \backslash \{(0, 0)\}$.

### A. Results

For any pmf $\psi$ over $\tilde{\mathcal{X}}$, define the quantity (50) on top of the next page. Define further

$$\mathbb{D}_Y(x_1, x_2) = \mathbb{D}\left(\Gamma_{Y|X_1X_2}(\cdot|x_1, x_2) \,\|\, \Gamma_{Y|X_1X_2}(\cdot|0, 0)\right), \tag{51}$$

$$\mathbb{D}_Z(x_1, x_2) = \mathbb{D}\left(\Gamma_{Z|X_1X_2}(\cdot|x_1, x_2) \,\|\, \Gamma_{Z|X_1X_2}(\cdot|0, 0)\right). \tag{52}$$

and the rectangular region $\mathcal{R}_\psi^\star(k_1, k_2)$ given by the nonnegative pairs $(r_1, r_2)$ satisfying Inequalities (53) on top of the next page, where we define $a/0 = \infty$.

*Theorem 5:* The covert capacity-key tradeoff region of the two-user DMMAC is

$$\mathcal{R}^\star(k_1, k_2) = \bigcup_{\psi} \mathcal{R}_\psi^\star(k_1, k_2), \tag{54}$$

where the union is taken over all pmfs $\psi$ over the alphabet $\tilde{\mathcal{X}}$.

*Proof:* See Appendix A. ■

*Remark 3 (Local Randomness only Required at One Transmitter):* Inspecting the achievability proof in Appendix A, we notice that local randomness $C_j$ is required only at most one of the Txs. In certain cases (for example for maximum rates at both users) no local randomness is required.

*Remark 4 (Opacity to channel transition probabilities for non-zero input pairs):* Above covert capacity-key tradeoff only depends on the channel transition probabilities $\Gamma_{Y|X_1X_2}(\cdot|x_1, x_2)$ and $\Gamma_{Z|X_1X_2}(\cdot|x_1, x_2)$ for input pairs $(x_1, x_2) \in \tilde{\mathcal{X}} \cup \{(0, 0)\}$ but not on the transition probabilities for other input pairs. In other words, only the output distributions matter corresponding to inputs where at least one of the two input symbols is non zero. This is in stark contrast to the

$$\chi^2(\psi) \triangleq \sum_{z\in\mathcal{Z}} \frac{\left(\sum_{(x_1,x_2)\in\tilde{\mathcal{X}}} \psi(x_1,x_2)\Gamma_{Z|X_1X_2}(z|x_1,x_2) - \Gamma_{Z|X_1X_2}(z|0,0)\right)^2}{\Gamma_{Z|X_1X_2}(z|0,0)}. \tag{50}$$

$$r_1 \leq \min\left\{\frac{\sum_{x_1\in\mathcal{X}_1\backslash\{0\}} \psi(x_1,0)\cdot\mathbb{D}_Y(x_1,0)\cdot k_1}{\max\left\{\sum_{x_1\in\mathcal{X}_1\backslash\{0\}} \psi(x_1,0)\cdot(\mathbb{D}_Z(x_1,0)-\mathbb{D}_Y(x_1,0)),\ 0\right\}},\ \sqrt{2}\frac{\sum_{x_1\in\mathcal{X}_1\backslash\{0\}} \psi(x_1,0)\cdot\mathbb{D}_Y(x_1,0)}{\sqrt{\chi^2(\psi)}}\right\} \tag{53a}$$

$$r_2 \leq \min\left\{\frac{\sum_{x_2\in\mathcal{X}_2\backslash\{0\}} \psi(0,x_2)\cdot\mathbb{D}_Y(0,x_2)\cdot k_2}{\max\left\{\sum_{x_2\in\mathcal{X}_2\backslash\{0\}} \psi(0,x_2)\cdot(\mathbb{D}_Z(0,x_2)-\mathbb{D}_Y(0,x_2)),\ 0\right\}},\ \sqrt{2}\frac{\sum_{x_2\in\mathcal{X}_2\backslash\{0\}} \psi(0,x_2)\cdot\mathbb{D}_Y(0,x_2)}{\sqrt{\chi^2(\psi)}}\right\}. \tag{53b}$$

traditional (non-covert) capacity region of a DMMAC, which generally depends on all conditional transition probabilities.

Our Theorem 5 simplifies for binary inputs, e.g., $\mathcal{X}_1 = \mathcal{X}_2 = \{0,1\}$, to:

*Corollary 6 (Binary Inputs):* For binary inputs, e.g., $\mathcal{X}_1 = \mathcal{X}_2 = \{0,1\}$, the covert capacity-key tradeoff region simplifies to:

$$\mathcal{C}(k_1,k_2) = \bigcup_{\rho\in[0,1]} \{0\leq r_1\leq r_1(\rho)\}\times\{0\leq r_2\leq r_2(\rho)\} \tag{55}$$

where

$$\rho_1(\rho) \triangleq \min\left\{\frac{\rho\mathbb{D}_Y(1,0)}{\max\{\rho(\mathbb{D}_Z(1,0)-\mathbb{D}_Y(1,0)),\ 0\}}k_1,\ \sqrt{2}\frac{\rho\mathbb{D}_Y(1,0)}{\sqrt{\chi^2(\rho)}}\right\} \tag{56}$$

$$\rho_2(\rho) \triangleq \min\left\{\frac{(1-\rho)\mathbb{D}_Y(0,1)}{\max\{(1-\rho)(\mathbb{D}_Z(0,1)-\mathbb{D}_Y(0,1)),\ 0\}}k_2,\ \sqrt{2}\frac{(1-\rho)\mathbb{D}_Y(0,1)}{\sqrt{\chi^2(\rho)}}\right\} \tag{57}$$

and $\chi^2(\rho)$ is defined in (58) on top of the next page.

### B. Discussion and Simplification of $\mathcal{R}^\star(k_1,k_2)$

In analogy to the single-user case, the structure of the cover capacity-key tradeoff region $\mathcal{R}^\star(k_1,k_2)$ depends on whether the differences

$$\mathbb{D}_Z(x_1,0)-\mathbb{D}_Y(x_1,0)>0, \qquad x_1\in\mathcal{X}_1\backslash\{0\}, \tag{59a}$$

$$\mathbb{D}_Z(0,x_2)-\mathbb{D}_Y(0,x_2)>0, \qquad x_2\in\mathcal{X}_2\backslash\{0\} \tag{59b}$$

are positive or not.

*Remark 5 (When no Key is Required at a Tx):* When (59a) is violated for all inputs $x_1\in\mathcal{X}_1\backslash\{0\}$, then the $r_1$-rate only depends on the choice of the pmf $\psi$ but not on $k_1$. Moreover, the extreme $r_1$-point of the covert capacity-key tradeoff region $\mathcal{R}^\star(k_1,k_2)$ neither depends on the key rate $k_2$; the region itself however does. Similar observations hold for $r_2$ when (59b) holds for all inputs $x_2\in\mathcal{X}_2\backslash\{0\}$.

Above remark sheds light on the behaviour when the adversary is uniformly *weaker* (over all inputs) than the legitimate Rx. The following corollary treats the opposite case where the adversary is uniformly *stronger* than the legitimate Rx.

*Corollary 7 (Adversary Stronger than Legitimate Receiver):* If (59) hold for all $x_1\in\mathcal{X}_1\backslash\{0\}$ and $x_2\in\mathcal{X}_2\backslash\{0\}$, then

$$\mathcal{R}^\star(k_1,k_2) = \begin{cases} [0,k_1S_{1,\max}]\times[0,k_2S_{2,\max}] & \text{if } k_1\leq k_{1,\text{lin}},\ k_2\leq k_{2,\text{lin}}, \\ \max\limits_{\psi\in\mathcal{L}(k_1,k_2)} \mathcal{R}^\star_\psi(k_1,k_2) & \text{if } k_1\geq k_{1,\text{lin}}, \text{ or } k_2\geq k_{2,\text{lin}}, \end{cases} \tag{60}$$

where

$$S_{1,\max} \triangleq \max_{x_1\in\mathcal{X}_1\backslash\{0\}} \frac{\mathbb{D}_Y(x_1,0)}{\mathbb{D}_Z(x_1,0)-\mathbb{D}_Y(x_1,0)}, \tag{61a}$$

$$S_{2,\max} \triangleq \max_{x_2\in\mathcal{X}_2\backslash\{0\}} \frac{\mathbb{D}_Y(0,x_2)}{\mathbb{D}_Z(0,x_2)-\mathbb{D}_Y(0,x_2)}, \tag{61b}$$

and

$$k_{1,\text{lin}} \triangleq \sqrt{2}\frac{\mathbb{D}_Z(x_{1,\text{best}},0)-\mathbb{D}_Y(x_{1,\text{best}},0)}{\sqrt{\chi^2(\delta_{x_1,0})}}, \tag{62}$$

$$k_{2,\text{lin}} \triangleq \sqrt{2}\frac{\mathbb{D}_Z(0,x_{2,\text{best}})-\mathbb{D}_Y(0,x_{2,\text{best}})}{\sqrt{\chi^2(\delta_{0,x_2})}}, \tag{63}$$

for $x_{1,\text{best}}$ and $x_{2,\text{best}}$ the maximizers in (61) and

$$\mathcal{L}(k_1,k_2) \triangleq \left\{\psi\colon\ k_1\geq\sqrt{2}\frac{\sum_{x_1\in\mathcal{X}_1\backslash\{0\}}\psi(x_1,0)\cdot\mathbb{D}_Y(x_1,0)}{\sqrt{\chi^2(\psi)}},\ k_2\geq\sqrt{2}\frac{\sum_{x_2\in\mathcal{X}_2\backslash\{0\}}\psi(0,x_2)\cdot\mathbb{D}_Y(0,x_2)}{\sqrt{\chi^2(\psi)}}\right\}. \tag{64}$$

*Proof:* Under Condition (59), irrespective of the pmf $\psi$, the first term in (53) is stringent for sufficiently small key values $k_1,k_2$ because the denominator is non-zero. Thus, for small key values the largest $r_1$ and $r_2$ values that are achievable with a given input pmf $\psi$ are given by the first expressions in the minimizations of (53). Notice further that these first expressions do not depend on the overall pmf $\psi$, but only on the $x_1$- and $x_2$- "conditionals", respectively. That means, the first term in (71a) remains unchanged if we multiply all $\{\psi(x_1,0)\}_{x_1}$ probabilities with a common normalization factor and similarly, the first term in (71b) remains unchanged if we multiply all $\{\psi(0,x_2)\}_{x_2}$ probabilities with a common normalization factor.

$$\chi^2(\rho) \triangleq \sum_{z\in\mathcal{Z}} \frac{\left(\rho\Gamma_{Z|X_1X_2}(z|1,0) + (1-\rho)\Gamma_{Z|X_1X_2}(z|0,1) - \Gamma_{Z|X_1X_2}(z|0,0)\right)^2}{\Gamma_{Z|X_1X_2}(z|0,0)} \tag{58}$$

For small key-values the two constraints can thus be optimized individually, each optimization being of the same flavour as in the single-user case where we proved that for small key values under assumption (23) (which corresponds to Assumption (59)) the optimal capacity-key tradeoff is achieved by a singleton distribution putting all mass on a non-zero value. Moreover, this pmf remains optimal until the corresponding capacity-key tradeoff saturates. We can employ similar arguments on each of the two dimensions of our two-dimension rate-region to conclude that the singleton distributions on the two inputs $x_{1,\text{best}} \in \mathcal{X}_1\backslash\{0\}$ and $x_{2,\text{best}} \in \mathcal{X}_2\backslash\{0\}$ achieve largest capacity-key tradeoff regions for all keys below $k_{1,\text{lin}}$ and $k_{2,\text{lin}}$, respectively. ■

*Remark 6 (Optimality of Binary Signaling for Strong Adversaries and Small Key Rates):* From above proof outline we deduce that binary signaling (i.e. $\mathcal{X}_j = \{0, x_{j,\text{best}}\}, \forall j \in \{1,2\}$) at both Txs is optimal when the adversary is stronger than the legitimate Rx and in the regime of small key rates. Moreover, in this regime there is no tradeoff between the largest covert rates that the two users can simultaneously achieve. For larger key rates however a tradeoff arises between the largest rates that are simultaneously achievable at the two users.

We end this subsection with numerical examples.

*Example 3 (Binary Inputs with a Strong Adversary):* Consider a DMMAC with binary input alphabets $\mathcal{X}_1 = \mathcal{X}_2 = \{0,1\}$, output alphabet $\mathcal{Y} = \mathcal{Z} = \{0,1,2,3\}$, and channel transition pmfs:

$$\Gamma_{Y|X_1X_2} = \begin{bmatrix} 0.25 & 0.16 & 0.08 & 0.51 \\ 0.45 & 0.10 & 0.35 & 0.10 \\ 0.04 & 0.18 & 0.20 & 0.58 \\ 0.15 & 0.18 & 0.32 & 0.35 \end{bmatrix} \tag{65a}$$

and

$$\Gamma_{Z|X_1X_2} = \begin{bmatrix} 0.24 & 0.39 & 0.32 & 0.05 \\ 0.49 & 0.03 & 0.41 & 0.07 \\ 0.22 & 0.39 & 0.23 & 0.16 \\ 0.25 & 0.10 & 0.35 & 0.30 \end{bmatrix}, \tag{65b}$$

where the four columns correspond to the four output symbols $y = 0,1,2,3$ and the four rows correspond to the four possible input pairs $(x_1, x_2)$ in increasing alphabetical ordering, i.e., $(0,0),(0,1),(1,0),(1,1)$. Notice that above channels satisfy Conditions (59) for the only non-zero inputs $x_1 = 1$ and $x_2 = 1$. This implies that the adversary is uniformly (over all inputs) stronger than the legitimate Rx (in KL-sense) and thus positive key rates are required to achieve positive covert rates.

Figures 5 and 6 illustrate the capacity-key tradeoff regions for this channel at different secret-key rates. We observe that for small key rates, the tradeoff regions are square regions and there is no tradeoff between the largest covert rates $r_1$ and $r_2$ that are simultaneously achievable at the two users. For larger key rates a tradeoff arises between the two rates

*Example 4 (Stronger Adversary):* Consider now a DMMAC with ternary input alphabets $\mathcal{X}_1 = \mathcal{X}_2 = \{0,1,2\}$ and let the output alphabets $\mathcal{Y} = \mathcal{Z} = \{0,1,2,3\}$. Assume that the channel transition laws are given by

$$\Gamma_{Y|X_1X_2} = \begin{bmatrix} 0.2 & 0.28 & 0.28 & 0.24 \\ 0.05 & 0.1 & 0.45 & 0.4 \\ 0.07 & 0.37 & 0.4 & 0.16 \\ 0.23 & 0.24 & 0.2 & 0.33 \\ 0.18 & 0.4 & 0.14 & 0.28 \\ 0.2 & 0.27 & 0.5 & 0.03 \\ 0.05 & 0.35 & 0.49 & 0.11 \\ 0.2 & 0.42 & 0.25 & 0.13 \\ 0.32 & 0.47 & 0.01 & 0.2 \end{bmatrix} \tag{66a}$$

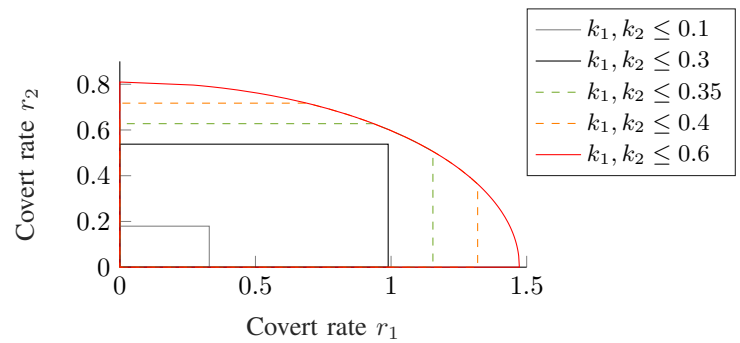


Fig. 5: $\mathcal{R}^*(k_1, k_2)$ for the channel in (65) when symmetric constraints are imposed on the key rates $k_1$ and $k_2$.

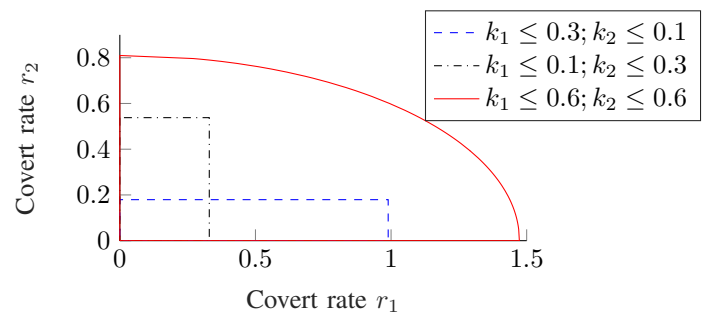


Fig. 6: $\mathcal{R}^*(k_1, k_2)$ for the channel in (65) for different symmetric and asymmetric constraints imposed on the key rates $k_1$ and $k_2$.

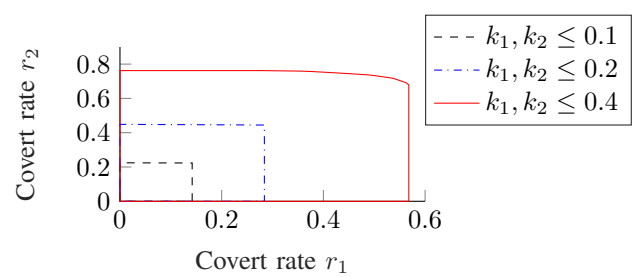


Fig. 7: $\mathcal{R}^*(k_1, k_2)$ for the channel in (66) under various symmetric constraints on the key rates $k_1$ and $k_2$.

and

$$\Gamma_{Z|X_1X_2} = \begin{bmatrix} 0.2 & 0.2 & 0.36 & 0.24 \\ 0.01 & 0.37 & 0.17 & 0.45 \\ 0.42 & 0.35 & 0.05 & 0.18 \\ 0.25 & 0.25 & 0.15 & 0.37 \\ 0.6 & 0.27 & 0.01 & 0.12 \\ 0.3 & 0.1 & 0.33 & 0.27 \\ 0.1 & 0.22 & 0.67 & 0.01 \\ 0.38 & 0.32 & 0.25 & 0.05 \\ 0.17 & 0.3 & 0.23 & 0.3 \end{bmatrix}, \tag{66b}$$

where the four columns correspond to the four output symbols $0, 1, 2, 3$ and the nine rows correspond to the nine distinct pairs $(x_1, x_2)$ in increasing alphabetical ordering, i.e., $(0,0), (0,1), (1,0), (1,1), \ldots, (2,2)$.

For this channel Conditions (59) are satisfied for all input pairs $(x_1, x_2) \in \tilde{\mathcal{X}}$. That means the adversary is again uniformly (over all inputs) stronger than the legitimate Rx (in KL-sense) and thus positive key rates are required to achieve positive covert rates. Figure 7 illustrates the capacity key-rate tradeoff regions for this channel at different secret-key rates. We observe that for small key rates, the tradeoff region is a square region and there is no tradeoff between the largest covert rates $r_1$ and $r_2$ that are simultaneously achievable at the two users. For larger key rates a tradeoff arises between the covert rates.

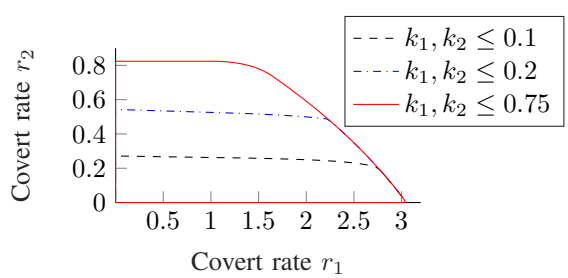


Fig. 8: $\mathcal{R}^\star(k_1, k_2)$ for the channel in (67) when symmetric constraints are imposed on the key rates $k_1$ and $k_2$.

*Example 5 (Mixed (Stronger/Weaker) Adversary ):* Consider another DMMAC with ternary input alphabets $\mathcal{X}_1 = \mathcal{X}_2 = \{0, 1, 2\}$, output alphabets $\mathcal{Y} = \mathcal{Z} = \{0, 1, 2, 3\}$, and channel transition laws

$$\Gamma_{Y|X_1X_2} = \begin{bmatrix} 0.20 & 0.29 & 0.28 & 0.23 \\ 0.05 & 0.10 & 0.44 & 0.41 \\ 0.07 & 0.37 & 0.40 & 0.16 \\ 0.31 & 0.47 & 0.01 & 0.21 \\ 0.23 & 0.24 & 0.21 & 0.33 \\ 0.18 & 0.4 & 0.14 & 0.29 \\ 0.21 & 0.27 & 0.50 & 0.02 \\ 0.04 & 0.35 & 0.50 & 0.11 \\ 0.21 & 0.42 & 0.25 & 0.12 \end{bmatrix} \tag{67a}$$

and

$$\Gamma_{Z|X_1X_2} = \begin{bmatrix} 0.20 & 0.19 & 0.36 & 0.25 \\ 0.01 & 0.37 & 0.17 & 0.45 \\ 0.42 & 0.35 & 0.05 & 0.18 \\ 0.17 & 0.31 & 0.22 & 0.30 \\ 0.25 & 0.25 & 0.19 & 0.31 \\ 0.60 & 0.27 & 0.02 & 0.11 \\ 0.29 & 0.10 & 0.33 & 0.28 \\ 0.10 & 0.22 & 0.67 & 0.01 \\ 0.38 & 0.32 & 0.25 & 0.05 \end{bmatrix}, \tag{67b}$$

For this channel, Condition (59b) is satisfied for all non-zero inputs $x_2$ but at the same time Condition (59a) is violated for all non-zero inputs $x_1$. That means, depending on the employed input distribution, the adversary can be weaker or stronger in detecting the communication than the legitimate Rx. As described in Remark 5, the covert capacity-key tradeoff region $\mathcal{R}^\star(k_1, k_2)$ does not depend on the key-rate $k_1$ but only on $k_2$. Figure 8 illustrates the capacity key-rate tradeoff regions for this channel at different key rates. We observe that even for small key rates there is a tradeoff between the simultaneously achievable covert rates $r_1$ and $r_2$.

### C. Generalization to $L \geq 2$ Transmitters

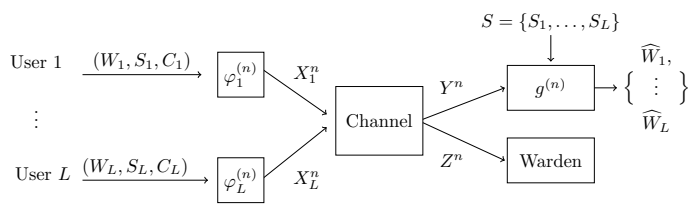


Fig. 9: Covert communication over a $L \geq 2$ -users MAC.

Consider now a general DMMAC with $L \geq 2$ Txs. Input alphabets are $\mathcal{X}_1, \ldots, \mathcal{X}_L$ and the channel is described by the transition laws $\Gamma_{Y|X_1 \cdots X_L}$ and $\Gamma_{Z|X_1 \cdots X_L}$. The setup is analogous as for the two-user case, in particular, see Figure 9. As for the two-user case, one can prove the following proposition for $L \geq 2$ users.

*Proposition 8 (General Number of $L \geq 2$ Transmitters):* The covert capacity-key tradeoff region is given by

$$\mathcal{R}^\star(k_1, \ldots, k_L) = \bigcup_{\psi} \mathcal{R}^\star_\psi(k_1, \ldots, k_L), \tag{68}$$

where the union is over pmfs over the set $\tilde{\mathcal{X}} = ((\mathcal{X}_1 \backslash \{0\}) \times \{0\}^{L-1}) \cup (\{0\} \times (\mathcal{X}_2 \backslash \{0\}) \times \{0\}^{L-2}) \cup \cdots \cup (\{0\}^{L-1} \times (\mathcal{X}_L \backslash \{0\}))$ and for a given $\psi$ the region $\mathcal{R}^\star_\psi(k_1, \ldots, k_L)$ consists of all nonnegative rate-tuples $(r_1, \ldots, r_L)$ satisfying Equation (69) on top of the next page, where $\mathbf{e}_\ell$ denotes the unit vector with a single non-zero component at position $\ell$.

*Proof:* Omitted. Analogous to the two-user case. ■

*Remark 7:* As for the two-user case, the region $\mathcal{R}^\star(k_1, \ldots, k_L)$ can be simplified when the adversary is stronger than the legitimate Rx (in the previously encountered KL sense) and key rates are low. In this case, it can again be shown that binary signaling is optimal for all Txs.

$$r_\ell \leq \min\left\{\frac{\sum_{x_\ell\in\mathcal{X}_\ell\setminus\{0\}}\psi(x_\ell\cdot\mathbf{e}_\ell)\cdot\mathbb{D}_Y(x_\ell\cdot\mathbf{e}_\ell)}{\max\left\{\sum_{x_\ell\in\mathcal{X}_\ell\setminus\{0\}}\psi(x_\ell\cdot\mathbf{e}_\ell)\cdot\left(\mathbb{D}_Z(x_\ell\cdot\mathbf{e}_\ell)-\mathbb{D}_Y(x_\ell\cdot\mathbf{e}_\ell)\right),\ 0\right\}}\cdot k_\ell,\ \sqrt{2}\frac{\sum_{x_\ell\in\mathcal{X}_\ell\setminus\{0\}}\psi(x_\ell\cdot\mathbf{e}_\ell)\cdot\mathbb{D}_Y(x_\ell\cdot\mathbf{e}_\ell)}{\sqrt{\chi^2(\psi)}}\right\}. \tag{69}$$

## IV. Discrete Memoryless Interference Channels

Naturally, our results also extend to the two-user (or multi-user) Discrete Memoryless Interference Channel (DMIC). We only present the setup and results here. The proofs and discussions are analogous to the DMMAC, just keeping in mind that there are two legitimate Rxs with different outputs $Y_1$ and $Y_2$.

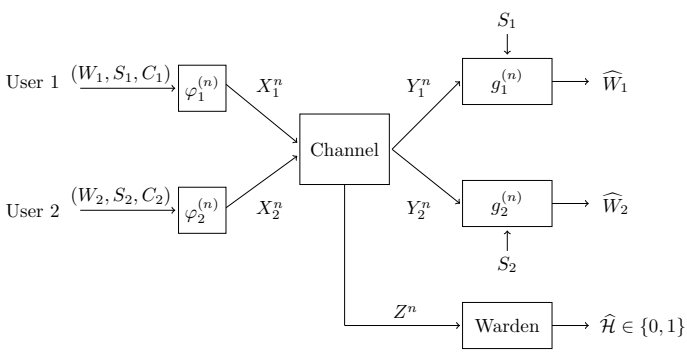


Fig. 10: Covert communication over a two-user DMIC.

Consider the setup in Figure 10, where notice that in contrast to the DMMAC we now have two legitimate Rxs observing different output sequences $Y_1^n$ and $Y_2^n$ and attempting to decode message $W_1$ and $W_2$, respectively. The mathematical description, covertness constraint, and definition of covert capacity-key tradeoff are analogous to the DMMAC.

*Theorem 9:* The covert capacity-key tradeoff region of the DMIC is:

$$\mathcal{R}^*_{\text{IC}}(k_1,k_2) = \bigcup_{\psi}\mathcal{R}^\star_{\text{IC},\psi}(k_1,k_2), \tag{70}$$

where the union is taken over all pmfs $\psi$ over the alphabet $\tilde{\mathcal{X}} = ((\mathcal{X}_1\backslash\{0\})\times\{0\})\cup(\{0\}\times(\mathcal{X}_2\backslash\{0\}))$ and for each $\psi$ the region $\mathcal{R}^\star_{\text{IC},\psi}(k_1,k_2)$ contains all non-negative rate-pairs $(r_1,r_2)$ satisfying Inequalities (71) on the next page where

$$\mathbb{D}_{Y_1}(x_1,0) \triangleq \mathbb{D}\left(\Gamma_{Y_1|X_1X_2}(\cdot|x_1,0)\,\|\,\Gamma_{Y_1|X_1X_2}(\cdot|0,0)\right) \tag{72}$$

$$\mathbb{D}_{Y_2}(0,x_2) \triangleq \mathbb{D}\left(\Gamma_{Y_2|X_1X_2}(\cdot|0,x_2)\,\|\,\Gamma_{Y_2|X_1X_2}(\cdot|0,0)\right). \tag{73}$$

*Proof:* Analogous to the MAC, but accounting for Rx $j\in\{1,2\}$ observing $Y_j^n$ and decoding $W_j$. ■

*Remark 8:* The region $\mathcal{R}^*_{\text{IC}}(k_1,k_2)$ can again be simplified when the adversary is stronger than the legitimate Rx (in the previously encountered KL sense) and key rates are low. In this case binary signaling is again optimal for both Txs.

## V. Summary and Conclusion

We determined the covert capacity-key tradeoff for single-user, multi-access and interference channels, i.e., the largest covert rates that are achievable for given key rates. Our results provide new insights into the relationships between the required key rates and the achievable covert rates. For single-user channels with binary inputs, the relationship between key-rate and largest achievable covert rate is linear for low key rates before it saturates at the covert capacity. For non-binary input channels the relationship is linear for small key rates only if the adversary is uniformly (over all nonzero inputs) stronger (in a Kullback-Leibler-divergence sense) than the legitimate Rx. In this case, binary signaling between the zero and a single non-zero input is optimal in the regime of small key rates. For larger key rates or if the adversary is not uniformly stronger than the legitimate Rx the largest achievable rate grows sublinearly in the key-rate before it saturates at the covert capacity. Binary signaling is not necessarily optimal in these cases.

In this work, we further provided first complete characterizations of the covert capacity-key tradeoff region for discrete memoryless multiple-access channels (DMMAC) with and without message cooperation, and for discrete memoryless interference channel (DMIC). Our results allowed us to shed light on these regions. In particular, we showed that similarly to the DMC, the behavior of these regions depends on whether the warden observes a stronger channel than the legitimate receiver for the different inputs. When the warden is consistently stronger over all inputs, binary signaling is optimal at low key rates, and no tradeoff exists between the users' covert rateseach scales linearly with its own key rate. At higher key rates, the tradeoff region grows sublinearly, introducing interdependencies between the users rates. When the warden is stronger for some inputs but weaker for others, the tradeoff region exhibits sublinear growth at all key rates and maintains an inherent rate tradeoff between users.

## Appendix A
## Proof of Theorem 5

We will first prove the following proposition.

*Proposition 10:* There exists a sequence (in the blocklength $n$) of tuples $(m_1,m_2,p_1,p_2,g_1,g_2)$ and encoding/decoding functions $(\varphi_1^{(n)},\varphi_2^{(n)},g^{(n)})$ satisfying (46)–(47) if, and only if, there exists a nonnegative number $\beta_1,\beta_2\in[0,1]$ and a pmf $\psi(\cdot)$ over $\tilde{\mathcal{X}}$ so that the key and message rates satisfy

$$r_1 \leq \beta_1\sqrt{2}\frac{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)\cdot\mathbb{D}_Y(x_1,0)}{\sqrt{\chi^2(\psi)}} \tag{74}$$

$$r_2 \leq \beta_2\sqrt{2}\frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi(0,x_2)\cdot\mathbb{D}_Y(0,x_2)}{\sqrt{\chi^2(\psi)}} \tag{75}$$

$$k_1 \geq \beta_1\sqrt{2}\frac{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)(\mathbb{D}_Z(x_1,0)-\mathbb{D}_Y(x_1,0))}{\sqrt{\chi^2(\psi)}} \tag{76}$$

$$k_2 \geq \beta_2\sqrt{2}\frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi(0,x_2)(\mathbb{D}_Z(0,x_2)-\mathbb{D}_Y(0,x_2))}{\sqrt{\chi^2(\psi)}}. \tag{77}$$

*Proof of Proposition 10:* We first sketch the achievability proof, then the converse.

$$r_1 \leq \min\left\{ \frac{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}} \psi(x_1,0)\cdot \mathbb{D}_{Y_1}(x_1,0)}{\max\left\{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}} \psi(x_1,0)\cdot\left(\mathbb{D}_Z(x_1,0)-\mathbb{D}_{Y_1}(x_1,0)\right),\ 0\right\}}\cdot k_1,\ \sqrt{2}\frac{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}} \psi(x_1,0)\cdot \mathbb{D}_{Y_1}(x_1,0)}{\sqrt{\chi^2(\psi)}}\right\} \tag{71a}$$

$$r_2 \leq \min\left\{ \frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}} \psi(0,x_2)\cdot \mathbb{D}_{Y_2}(0,x_2)}{\max\left\{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}} \psi(0,x_2)\cdot\left(\mathbb{D}_Z(0,x_2)-\mathbb{D}_{Y_2}(0,x_2)\right),\ 0\right\}}\cdot k_2,\ \sqrt{2}\frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}} \psi(0,x_2)\cdot \mathbb{D}_{Y_2}(0,x_2)}{\sqrt{\chi^2(\psi)}}\right\}. \tag{71b}$$

### A. Coding Scheme (Achievability Proof)

Assume that $\beta_1 \geq \beta_2$, otherwise switch the roles of the two users. Set $\phi_j = \beta_j^2$, for $j \in \{1,2\}$. In our coding scheme, Tx1 communicates during the first $n_1 \triangleq \lfloor \phi_1 n \rfloor$ channel uses and deterministically sends the all-zero sequence during the last $n - n_1$ channel uses. Tx 2 communicates only during the first $n_2 \triangleq \lfloor \phi_2 n \rfloor \leq n_1$ channel uses; during the following $n_1 - n_2$ channel uses it sends completely random symbols according to a distribution that we shall specify shortly; and during the remaining $n - n_1$ channel uses it sends the all-zero sequence. In our scheme thus only Tx 2 uses its local randomness and only if $\beta_1 > \beta_2$. Similarly, if $\beta_2 > \beta_1$ then Tx 1 uses its local randomness. Figure 11 illustrates this scheme. We next explain how the codewords $x_1^{n_1}(W_1,S_1)$ and $x_2^{n_2}(W_2,S_2)$ transmitted during the first phases are generated, as well as the random inputs generated at Tx 2.

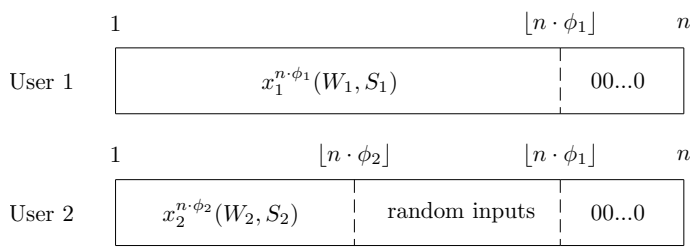


Fig. 11: Covert coding scheme for the MAC.

As in [13], we pick a sequence $\{\omega_n\}_{n=1}^{\infty}$ satisfying

$$\lim_{n\to\infty} \omega_n = 0, \tag{78a}$$

$$\lim_{n\to\infty} \left(\omega_n\sqrt{n} - \log n\right) = \infty, \tag{78b}$$

and define

$$\alpha_n \triangleq \frac{\omega_n}{\sqrt{n}}. \tag{79}$$

We notice that $\alpha_n$ vanishes as $n \to \infty$, but $\alpha_n \cdot n$ grows to infinity.

We further pick a pmf $\psi$ over the previously defined set $\tilde{\mathcal{X}} = ((\mathcal{X}_1\setminus\{0\}) \times \{0\}) \cup (\{0\}\times(\mathcal{X}_1\setminus\{0\}))$ and define the pmfs

$$P_{X_{1,n}}(x_1) = \psi(x_1,0)\cdot\alpha_n, \quad x_1 \in \mathcal{X}_1\setminus\{0\}, \tag{80}$$

$$P_{X_{2,n}}(x_2) = \psi(0,x_2)\cdot\alpha_n, \quad x_2 \in \mathcal{X}_2\setminus\{0\}, \tag{81}$$

$$P_{X_{j,n}}(0) = 1 - \sum_{x_j\in\mathcal{X}_j\setminus\{0\}} P_{X_{j,n}}(x_j), \quad j\in\{1,2\}, \tag{82}$$

$$P^{(n)}_{X_1X_2Y}(x_1,x_2,y) \triangleq P_{X_{1,n}}(x_1)P_{X_{2,n}}(x_2)\Gamma_{Y|X_1X_2}(y|x_1,x_2), \quad \forall (x_1,x_2,y)\in\mathcal{X}_1\times\mathcal{X}_2\times\mathcal{Y}. \tag{83}$$

*Codebook generation:*
For each $i = 1,\dots,n_1$, draw the $i$-th entry of each codeword $x_1^{n_1}(w_1,s_1)$ according to the pmf $P_{X_{1,n}}$ independent of all other entries.

For each $i = 1,\dots,n_2$, draw the $i$-th entry of each codeword $x_2^{n_2}(w_2,s_2)$ according to the pmf $P_{X_{2,n}}$ independent of all other entries.

The codebooks are revealed to all parties.

*Decoding at the Receiver:* Choose an arbitrary small $\mu > 0$. For any blocklength $n'$ and positive constant $\eta$, define the set

$$\mathcal{A}_\eta^{n'} \triangleq \left\{ (x_1^{n'}, x_2^{n'}, y^{n'}) \colon \log\left(\frac{\Gamma^{\otimes n'}_{Y|X_1X_2}(y^{n'}|x_1^{n'},x_2^{n'})}{\Gamma^{\otimes n'}_{Y|X_1X_2}(y^{n'}|0^{n'},0^{n'})}\right) \geq \eta \right\}. \tag{84}$$

The legitimate Rx observes $Y^n = y^n$ and knows the secret-keys $(S_1,S_2)$. To decode message $W_1$, it looks for a unique index $w_1$ satisfying

$$(x_1^{n_1}(w_1,S_1), 0^{n_1}, y^{n_1}) \in \mathcal{A}_{\eta_1}^{n_1}, \tag{85}$$

where for any $n' < n$ we let $y^{n'}$ denote the first $n'$ symbols of $y^n$ and

$$\eta_1 \triangleq (1-\mu)\phi_1\sqrt{n}\omega_n \left(\sum_{x_1\in\mathcal{X}_1\setminus\{0\}} \psi(x_1,0)\mathbb{D}_Y(x_1,0)\right). \tag{86}$$

If such a unique index $w_1$ exists, the legitimate Rx sets $\widehat{W_1} = w_1$. Otherwise, it declares an error and stops.

Similarly, to decode message $W_2$, the legitimate Rx looks for a unique index $w_2$ satisfying

$$(0^{n_2}, x_2^{n_2}(w_2,S_2), y^{n_2}) \in \mathcal{A}_{\eta_2}^{n_2}, \tag{87}$$

for

$$\eta_2 \triangleq (1-\mu)\phi_2\sqrt{n}\omega_n \left(\sum_{x_2\in\mathcal{X}_2\setminus\{0\}} \psi(0,x_2)\mathbb{D}_Y(0,x_2)\right). \tag{88}$$

If such a unique index $w_2$ exists, it sets $\widehat{W_2} = w_2$. Otherwise, it declares an error.

*Performance Analysis:* The first analysis steps are standard for covert communication and can be obtained, for example, by converting the steps in [16, Section IV] to a MAC where message $W_1$ is decoded based on the key and the first $n_1$ outputs of $Y^n$ and message $W_2$ based on the keys and the first $n_2$ outputs of $Y^n$. By these analysis steps, we obtain that for *any* small $\mu > 0$ and sufficiently large blocklengths $n$, we can choose

$$m_1 = (1-\mu)\phi_1\cdot\omega_n\sqrt{n}\sum_{x_1\in\mathcal{X}_1\setminus\{0\}} \psi(x_1,0)\cdot\mathbb{D}_Y(x_1,0), \tag{89a}$$

$$m_2 = (1-\mu)\phi_2 \cdot \omega_n \sqrt{n} \sum_{x_2 \in \mathcal{X}_2 \setminus \{0\}} \psi(0, x_2) \cdot \mathbb{D}_Y(0, x_2), \quad (89b)$$

while guaranteeing

$$\Pr[\widehat{W}_j \neq W_j] \leq e^{-B\phi_j \cdot \omega_n \sqrt{n}}, \quad j \in \{1, 2\}, \quad (90)$$

for a positive real number $B$.

The resolvability analysis requires more care. Recall the definition of the warden's output distribution $\widehat{Q}(z^n)$ in (44) and that the covertness constraint is measured as:

$$\delta_n = \mathbb{D}(\widehat{Q}^n \| \Gamma^{\otimes n}_{Z|X_1X_2}(\cdot|0^n, 0^n)). \quad (91)$$

Define the pmfs

$$\Gamma_{Z|X_1}(z|x_1) \triangleq \sum_{x_2 \in \mathcal{X}_2} P_{X_{2,n}}(x_2) \Gamma_{Z|X_1X_2}(z|x_1, x_2), \quad z \in \mathcal{Z}, x_1 \in \mathcal{X}_1, \quad (92)$$

$$\Gamma_{Z|X_1}(z|x_2) \triangleq \sum_{x_1 \in \mathcal{X}_1} P_{X_{1,n}}(x_1) \Gamma_{Z|X_1X_2}(z|x_1, x_2), \quad z \in \mathcal{Z}, x_2 \in \mathcal{X}_2, \quad (93)$$

and

$$\tilde{Q}(z) \triangleq \sum_{x_1 \in \mathcal{X}_1} \sum_{x_2 \in \mathcal{X}_2} P_{X_{1,n}}(x_1) P_{X_{2,n}}(x_2) \Gamma_{Z|X_1X_2}(z|x_1, x_2), \quad z \in \mathcal{Z}. \quad (94)$$

Notice that the covertness constraint term $\delta_n$ only depends on the first $n_1$ channel uses, as the terms corresponding to the last $n - n_1$ channel uses are zero. By Pinsker's inequality, we then have following bound holds:

$$\begin{aligned} &\left| \delta_n - \mathbb{D}\left(\tilde{Q}^{\otimes n_1} \,\|\, \Gamma^{\otimes n_1}_{Z|X_1X_2}(\cdot|0^{n_1}, 0^{n_1})\right) \right| \\ &\leq \mathbb{D}\left(\widehat{Q}^{n_1} \,\|\, \tilde{Q}^{\otimes n_1}\right) \\ &\quad + n_1 \log\left(\frac{1}{\min_{z \in \mathcal{Z}} \Gamma_{Z|X_1X_2}(z|0,0)}\right) \sqrt{\frac{1}{2}\mathbb{D}\left(\widehat{Q}^{n_1} \,\|\, \tilde{Q}^{\otimes n_1}\right)}, \end{aligned} \quad (95)$$

where $\widehat{Q}^{n_1}$ denotes the pmf of the warden's first $n_1$ output symbols .

Since $\alpha_n \to 0$ as $n \to \infty$, by [13, Eq. 295], we can write

$$\begin{aligned} &\mathbb{D}\left(\tilde{Q}^{\otimes n_1} \,\|\, \Gamma^{\otimes n_1}_{Z|X_1X_2}(\cdot|0^{n_1}, 0^{n_1})\right) \\ &= (1 + o(1)) \cdot \frac{n_1}{n} \cdot \frac{\omega_n^2}{2} \cdot \chi^2(\psi), \end{aligned} \quad (96)$$

where the final step holds by the definition of $\alpha_n$ in (79) and

$$n_1 \cdot \alpha_n^2 = n_1 \cdot \frac{\omega_n^2}{n}. \quad (97)$$

Since the ratio $\frac{n_1}{n} \to \phi_1$ and $\omega_n \to 0$, the terms in (96) vanish in the limit of large blocklengths $n$. However, they vanish slowlier than exponentially.

We wish show that the divergence term in the RHS of (95) does vanish exponentially in the blocklength, from which we conclude that the asymptotic scaling of $\delta_n$ is as given in (96).

To prove the desired statement, we will show that the average over all random codebooks $\mathcal{C}$ of the divergence $\mathbb{D}\left(\widehat{Q}^{n_1} \,\|\, \tilde{Q}^{\otimes n_1}\right)$ is vanishing exponentially fast. To make the dependence of the divergence on the choice of the random codebook more explicit, we introduce the codebook as a subscript of the output distribution. Then, defining

$$\widehat{Q}^{n_1}_{\mathcal{C}}(z^{n_1}) \triangleq \frac{1}{2^{m_1+m_2+p_1+p_2}} \sum_{(w_1,s_1)} \sum_{(w_2,s_2)} \widehat{Q}^{n_1}_{\mathcal{C},w_1,w_2,s_1,s_2}(z^{n_1}), \quad (98)$$

where for any valid $(w_1, w_2, s_1, s_2)$, we have:

$$\begin{aligned} \widehat{Q}^{n_1}_{\mathcal{C},w_1,w_2,s_1,s_2}(z^{n_1}) \triangleq \\ &\Gamma^{\otimes n_2}_{Z|X_1X_2}(z^{n_2}|x_1^{n_2}(w_1,s_1), x_2^{n_2}(w_2,s_2)) \\ &\cdot \Gamma^{(n_2 \to n_1)}_{Z|X_1}(z^{n_1}_{n_2+1}|x^{n_1}_{1,n_2+1}(w_1,s_1)), \end{aligned} \quad (99)$$

for $z^{n_1}_{n_2+1} \triangleq (z_{n_2+1}, \ldots, z_{n_1})$ and for $\Gamma^{(n_2 \to n_1)}_{Z|X_1}$ defined as:

$$\Gamma^{(n_2 \to n_1)}_{Z|X_1}(z^{n_1}_{n_2+1}|x^{n_1}_{1,n_2+1}(w_1,s_1)) \triangleq \prod_{i=n_2+1}^{n_1} \Gamma_{Z|X_1}(z_i|x_{1,i}). \quad (100)$$

We can now the sequence of equalities in (101) on top of the next page, where in $(a)$ we used that

$$\mathop{\mathbb{E}}_{\substack{X_1^{n_1}(w_1,s_1) \\ X_2^{n_2}(w_2,s_2)}} \left[\widehat{Q}^{n_1}_{\mathcal{C},w_1,w_2,s_1,s_2}(z^{n_1})\right] = \tilde{Q}^{\otimes n_1}(z^{n_1}) \quad (102)$$

and

$$\begin{aligned} \mathop{\mathbb{E}}_{X_1^{n_1}(w_1,s_1)} \left[\widehat{Q}^{n_1}_{\mathcal{C},w_1,1,s_1,1}(z^{n_1})\right] &= \Gamma^{\otimes n_2}_{Z|X_2}(z^{n_2}|X_2^{n_2}(1,1)) \\ &\quad \cdot \tilde{Q}^{\otimes(n_1-n_2)}(z^{n_1}_{n_2+1}) \end{aligned} \quad (103)$$

$$\mathop{\mathbb{E}}_{X_2^{n_2}(w_2,s_2)} \left[\widehat{Q}^{n_1}_{\mathcal{C},1,w_2,1,s_2}(z^{n_1})\right] = \Gamma^{\otimes n_1}_{Z|X_1}(z^{n_1}|X_1^{n_1}(1,1)) \quad (104)$$

and we simplified the last fraction by noting that

$$\begin{aligned} &\frac{\Gamma^{\otimes n_2}_{Z|X_2}(Z^{n_2}|X_2^{n_2}(1,1)) \cdot \tilde{Q}^{\otimes(n_1-n_2-1)}(Z^{n_1}_{n_2+1})}{\tilde{Q}^{\otimes n_1}(Z^{n_1})} \\ &= \frac{\Gamma^{\otimes n_2}_{Z|X_2}(Z^{n_2}|X_2^{n_2}(1,1))}{\tilde{Q}^{\otimes n_2}(Z^{n_1})}. \end{aligned} \quad (105)$$

Define now for any triple $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)$ the set:

$$\begin{aligned} &\mathcal{B}^{n_1}_{\boldsymbol{\theta}} \\ &\triangleq \Bigg\{ (x_1^{n_1}, x_2^{n_2}, z^{n_1}) \in \mathcal{X}_1^{n_1} \times \mathcal{X}_2^{n_2} \times \mathcal{Z}^{n_1} : \\ &\quad \log\left(\frac{\Gamma^{\otimes n_2}_{Z|X_1X_2}(z^{n_2}|x_1^{n_2}, x_2^{n_2}) \cdot \Gamma^{n_2 \to n_1}_{Z|X_1}(z^{n_1}_{n_2+1}|x^{n_1}_{1,n_2+1})}{\Gamma^{\otimes n_1}_{Z|X_1X_2}(z^{n_1}|0^{n_1}, 0^{n_1})}\right) \\ &\qquad \leq \theta_0, \\ &\quad \log\left(\frac{\Gamma^{n_1}_{Z|X_1}(z^{n_1}|x_1^{n_1})}{\Gamma^{\otimes n_1}_{Z|X_1X_2}(z^{n_1}|0^{n_1}, 0^{n_1})}\right) \leq \theta_1, \\ &\quad \log\left(\frac{\Gamma^{n_2}_{Z|X_2}(z^{n_2}|x_2^{n_2})}{\Gamma^{\otimes n_2}_{Z|X_1X_2}(z^{n_2}|0^{n_2}, 0^{n_2})}\right) \leq \theta_2 \Bigg\}, \end{aligned} \quad (106)$$

and notice that $x_2^{n_2}$ is of length $n_2$ while $x_1^{n_1}$ and $z_1^{n_1}$ are of length $n_1$. We then define the events

$$A = \{(X_1^{n_1}(W_1, S_1), X_2^{n_2}(W_2, S_2), Z^{n_1}) \in \mathcal{B}^{n_1}_{\boldsymbol{\theta}}\} \quad (107)$$

$$B = \{(X_1^{n_1}(W_1, S_1), X_2^{n_2}(W_2, S_2), Z^{n_1}) \notin \mathcal{B}^{n_1}_{\boldsymbol{\theta}}\} \quad (108)$$

$$\mathbb{E}_{\mathcal{C}}\left[\mathbb{D}\left(\widehat{Q}_{\mathcal{C}}^{n_1}\middle\|\tilde{Q}^{\otimes n_1}\right)\right]$$

$$= \mathbb{E}\left[\log\left(\sum_{(w_1,w_2,s_1,s_2)}\mathbb{E}_{\substack{\{X_1^{n_1}(w_1,s_1)\}\setminus X_1^{n_1}(1,1),\\ \{X_2^{n_2}(w_2,s_2)\}\setminus X_2^{n_2}(1,1)}}\left[\frac{\widehat{Q}_{\mathcal{C},w_1,w_2,s_1,s_2}^{n_1}(Z^{n_1})}{\mathsf{M}_1\mathsf{M}_2\mathsf{K}_1\mathsf{K}_2\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}\right]\right)\right] \tag{101a}$$

$$\leq \mathbb{E}\Bigg[\log\Bigg(\sum_{\substack{(w_1,s_1)\neq(1,1)\\(w_2,s_2)\neq(1,1)}}\mathbb{E}_{\substack{X_1^{n_1}(w_1,s_1)\\X_2^{n_2}(w_2,s_2)}}\left[\frac{\widehat{Q}_{\mathcal{C},w_1,w_2,s_1,s_2}^{n_1}(Z^{n_1})}{\mathsf{M}_1\mathsf{M}_2\mathsf{K}_1\mathsf{K}_2\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}\right]+\frac{\widehat{Q}_{\mathcal{C},1,1,1,1}^{n_1}(Z^{n_1})}{\mathsf{M}_1\mathsf{M}_2\mathsf{K}_1\mathsf{K}_2\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}$$

$$+\sum_{(w_2,s_2)\neq(1,1)}\mathbb{E}_{X_2^{n_2}(w_2,s_2)}\left[\frac{\widehat{Q}_{\mathcal{C},1,w_2,1,s_2}^{n_1}(Z^{n_1})}{\mathsf{M}_1\mathsf{M}_2\mathsf{K}_1\mathsf{K}_2\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}\right]$$

$$+\sum_{(w_1,s_1)\neq(1,1)}\mathbb{E}_{X_1^{n_1}(w_1,s_1)}\left[\frac{\widehat{Q}_{\mathcal{C},w_1,1,s_1,1}^{n_1}(Z^{n_1})}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}\right]\Bigg)\Bigg] \tag{101b}$$

$$\overset{(a)}{=}\mathbb{E}\Bigg[\log\Bigg(\frac{(2^{m_1+p_1}-1)(2^{m_2+p_2}-1)}{2^{m_1+m_2+p_1+p_2}}+\frac{\widehat{Q}_{\mathcal{C},1,1,1,1}^{n_1}(Z^{n_1})}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}$$

$$+\frac{(2^{m_2+p_2}-1)\Gamma_{Z|X_1}^{\otimes n_1}(z^{n_1}|X_1^{n_1}(1,1))}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{(\mathsf{M}_1\mathsf{K}_1-1)\Gamma_{Z|X_2}^{\otimes n_2}(z^{n_2}|X_2^{n_2}(1,1))}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_2}(Z^{n_2})}\Bigg)\Bigg] \tag{101c}$$

$$\leq \mathbb{E}\left[\log\left(1+\frac{\widehat{Q}_{\mathcal{C},1,1,1,1}^{n_1}(Z^{n_1})}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_1}^{n_1}(Z^{n_1}|X_1^{n_1}(1,1))}{2^{m_1+p_1}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_2}^{n_2}(Z^{n_2}|X_2^{n_2}(1,1))}{2^{m_2+p_2}\cdot\tilde{Q}^{\otimes n_2}(Z^{n_2})}\right)\right] \tag{101d}$$

and obtain by the total law of expectation, the inequality in (109) on top of next page holds.

We start by bounding the first summand as shown in (110) on top of next page, where $(a)$ holds by the definition of the set $\mathcal{B}_{\boldsymbol{\theta}}^n$ in (106); and $(b)$ holds because $Z^{n'}\sim\tilde{Q}^{\otimes n'}$ for any $n'$ and because pmfs sum to 1.

To bound the second summand in (109a), we notice that by definition, see (94), $\tilde{Q}(z)$ is lower bounded by $P_{X_{1,n}}(0)P_{X_{2,n}}(0)\min_{z\in\mathcal{Z}}\Gamma_{Z|X_1X_2}(z|0,0)$. Therefore, we can trivially bound the conditional expectation as shown in (111) on top of next page. To bound the probability of event $B$, we first notice the asymptotic behaviours of the following expectations (expectations are with respect to the codewords), see Inequality (112) on top of next page. where $\mathcal{O}(\alpha_n^2)$ denotes a function that vanishes in the order of $\alpha_n^2$.

Since also, the variances of above log-expressions vanish in the order of $\alpha_n^2$, we can apply Bernstein's Inequality and the union bound to conclude that if we set

$$\theta_{1,n}\triangleq(1+\xi_1)\alpha_n\Bigg(n_1\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)\mathbb{D}_Z(x_1,0)+n_2\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi(0,x_2)\mathbb{D}_Z(0,x_2)\Bigg), \tag{113a}$$

$$\theta_{2,n}\triangleq(1+\xi_2)\alpha_n\left(n_1\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)\mathbb{D}_Z(x_1,0)\right), \tag{113b}$$

$$\theta_{3,n}\triangleq(1+\xi_3)\alpha_n\left(n_2\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi(0,x_2)\mathbb{D}_Z(0,x_2)\right), \tag{113c}$$

for arbitrary small positive numbers $\xi_1,\xi_2,\xi_3$, then there exist positive constants $B_1,B_2,B_3>0$ such that

$$\Pr[B]\leq e^{-B_1\alpha_n n}+e^{-B_2\alpha_n n}+e^{-B_3\alpha_n n} \tag{114}$$

Combining (109a) with (110c), (111a), and (114), allows us to conclude that when

$$\limsup_{n\to\infty}(m_1+m_2+p_1+p_2-\theta_{1,n})=-\infty, \tag{115a}$$

$$\limsup_{n\to\infty}(m_1+p_1-\theta_{2,n})=-\infty, \tag{115b}$$

$$\limsup_{n\to\infty}(m_2+p_2-\theta_{3,n})=-\infty, \tag{115c}$$

then the divergence on the right-hand side of (95) tends to 0 exponentially fast in $\alpha_n n$, making the approximation

$$\mathbb{D}\left(\tilde{Q}^{\otimes n_1}\,\|\,\Gamma_{Z|X_1X_2}^{\otimes n_1}(\cdot|0^{n_1},0^{n_1})\right)=(1+o(1))\cdot\frac{n_1}{n}\cdot\frac{\omega_n^2}{2}\cdot\chi^2(\psi) \tag{116}$$

exponentially tight (in $\alpha_n n$). Combining further with (95) and (96), we conclude that under Conditions (115):

$$\delta_n=(1+o(1))\cdot\frac{n_1}{n}\cdot\frac{\omega_n^2}{2}\cdot\chi^2(\psi). \tag{117}$$

Using the expressions for the message bits in (89), and letting $n\to\infty$ and $\mu,\xi_1,\xi_2,\xi_3\downarrow 0$ (since t $n_1\approx n\phi_1$ and $n_2\approx n\phi_2$

$$\begin{aligned}
&\mathbb{E}_{\mathcal{C}}\left[\mathbb{D}\left(\widehat{Q}_{\mathcal{C}}^{n_1}\middle\|\tilde{Q}^{\otimes n_1}\right)\right]\\
&\leq \mathbb{E}\left[\log\left(1+\frac{\widehat{Q}_{\mathcal{C},1,1,1,1,1}^{n_1}(Z^{n_1})}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_1}^{n_1}(Z^{n_1}|X_1^{n_1}(1,1))}{2^{m_1+p_1}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_2}^{n_2}(Z^{n_2}|X_2^{n_2}(1,1))}{2^{m_2+p_2}\cdot\tilde{Q}^{\otimes n_2}(Z^{n_2})}\right)\middle|A\right]\cdot\Pr[A]\\
&\quad+\mathbb{E}\left[\log\left(1+\frac{\widehat{Q}_{\mathcal{C},1,1,1,1,1}^{n_1}(Z^{n_1})}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_1}^{n_1}(Z^{n_1}|X_1^{n_1}(1,1))}{2^{m_1+p_1}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_2}^{n_2}(Z^{n_2}|X_2^{n_2}(1,1))}{2^{m_2+p_2}\cdot\tilde{Q}^{\otimes n_2}(Z^{n_2})}\right)\middle|B\right]\cdot\Pr[B].
\end{aligned}\tag{109a}$$

$$\begin{aligned}
&\mathbb{E}\left[\log\left(1+\frac{\widehat{Q}_{\mathcal{C},1,1,1,1,1}^{n_1}(Z^{n_1})}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_1}^{n_1}(Z^{n_1}|X_1^{n_1}(1,1))}{2^{m_1+p_1}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_2}^{n_2}(Z^{n_2}|X_2^{n_2}(1,1))}{2^{m_2+p_2}\cdot\tilde{Q}^{\otimes n_2}(Z^{n_2})}\right)\middle|A\right]\cdot\Pr[A]\\
&\overset{(a)}{\leq}\mathbb{E}\left[\log\left(1+\frac{2^{\theta_0}\Gamma_{Z|X_1X_2}^{\otimes n_1}(Z^{n_1}|0^{n_1},0^{n_1})}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1}))}+\frac{2^{\theta_1}\Gamma_{Z|X_1X_2}^{\otimes n_1}(Z^{n_1}|0^{n_1},0^{n_1})}{2^{m_1+p_1}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{2^{\theta_2}\Gamma_{Z|X_1X_2}^{\otimes n_2}(Z^{n_2}|0^{n_2},0^{n_2})}{2^{m_2+p_2}\cdot\tilde{Q}^{\otimes n_2}(Z^{n_2})}\right)\middle|A\right]\cdot\Pr[A]
\end{aligned}\tag{110a}$$

$$\begin{aligned}
&\leq\frac{2^{\theta_0}}{2^{m_1+m_2+p_1+p_2}}\,\mathbb{E}\left[\frac{\Gamma_{Z|X_1X_2}^{\otimes n_1}(Z^{n_1}|0^{n_1},0^{n_1})}{\tilde{Q}^{\otimes n_1}(Z^{n_1})}\right]+\frac{2^{\theta_1}}{2^{m_1+p_1}}\,\mathbb{E}\left[\frac{\Gamma_{Z|X_1X_2}^{\otimes n_1}(Z^{n_1}|0^{n_1},0^{n_1})}{\tilde{Q}^{\otimes n_1}(Z^{n_1})}\right]\\
&\quad+\frac{2^{\theta_2}}{2^{m_2+p_2}}\,\mathbb{E}\left[\frac{\Gamma_{Z|X_1X_2}^{\otimes n_2}(Z^{n_2}|0^{n_2},0^{n_2})}{\tilde{Q}^{\otimes n_2}(Z^{n_2})}\right]
\end{aligned}\tag{110b}$$

$$\overset{(b)}{=}\frac{2^{\theta_0}}{2^{m_1+m_2+p_1+p_2}}+\frac{2^{\theta_1}}{2^{m_1+p_1}}+\frac{2^{\theta_2}}{2^{m_2+p_2}},\tag{110c}$$

$$\begin{aligned}
&\mathbb{E}\left[\log\left(1+\frac{\widehat{Q}_{\mathcal{C},1,1,1,1,1}^{n_1}(Z^{n_1})}{2^{m_1+m_2+p_1+p_2}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_1}^{n_1}(Z^{n_1}|X_1^{n_1}(1,1))}{2^{m_1+p_1}\cdot\tilde{Q}^{\otimes n_1}(Z^{n_1})}+\frac{\Gamma_{Z|X_2}^{n_2}(Z^{n_2}|X_2^{n_2}(1,1))}{2^{m_2+p_2}\cdot\tilde{Q}^{\otimes n_2}(Z^{n_2})}\right)\middle|B\right]\\
&\leq n_1\log\left(\frac{4}{\left(1-\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)\alpha_n\right)\left(1-\sum_{x2\in\mathcal{X}_2\setminus\{0\}}\psi(0,x_2)\alpha_n\right)\cdot\min_{z\in\mathcal{Z}}\Gamma_{Z|X_1X_2}(z|0,0)}\right).
\end{aligned}\tag{111a}$$

$$\begin{aligned}
&\mathbb{E}\left[\log\left(\frac{\Gamma_{Z|X_1X_2}^{\otimes n_2}(Z^{n_2}|X_1^{n_2}(1,1),X_2^{n_2}(1,1))\cdot\Gamma_{Z|X_1}^{n_2\to n_1}(Z_{n_2+1}^{n_1}|X_{1,n_2+1}^{n_1}(1,1))}{\Gamma_{Z|X_1X_2}^{\otimes n_1}(Z^{n_1}|0^{n_1},0^{n_1})}\right)\right]\\
&\leq\alpha_n\left(n_2\sum_{(x_1,x_2)\in\tilde{\mathcal{X}}}\psi(x_1,x_2)\mathbb{D}_Z(x_1,x_2)+(n_1-n_2)\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)\mathbb{D}_Z(x_1,0))\right)+n\mathcal{O}(\alpha_n^2)
\end{aligned}\tag{112a}$$

$$\begin{aligned}
&\mathbb{E}\left[\log\left(\frac{\Gamma_{Z|X_1}^{\otimes n_1}(Z^{n_1}|X_1^{n_1}(1,1))}{\Gamma_{Z|X_1X_2}^{\otimes n_1}(Z^{n_1}|0^{n_1},0^{n_1})}\right)\right]\\
&\leq\alpha_n n_1\cdot\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)\mathbb{D}_Z(x_1,0)+\mathcal{O}(\alpha_n^2)
\end{aligned}\tag{112b}$$

$$\begin{aligned}
&\mathbb{E}\left[\log\left(\frac{\Gamma_{Z|X_2}^{n_2}(Z^{\otimes n_2}|X_2^{n_2}(1,1))}{\Gamma_{Z|X_1X_2}^{\otimes n_2}(Z^{n_2}|0^{n_2},0^{n_2})}\right)\right]\\
&\leq\alpha_n n_2\cdot\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi(0,x_2)\mathbb{D}_Z(0,x_2)+\mathcal{O}(\alpha_n^2),
\end{aligned}\tag{112c}$$

and $\alpha_n=\omega_n\sqrt{n}$), we conclude achievability of the message-key rate tuples $(r_1,r_2,k_1,k_2)$

$$r_1=\sqrt{\phi_1}2\frac{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)\mathbb{D}_Y(x_1,0)}{\sqrt{\chi^2(\psi)}},\tag{118}$$

$$r_2 = \frac{\phi_1\sqrt{2}}{\sqrt{\phi_2}} \frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}} \psi(0,x_2)\mathbb{D}_Y(0,x_2)}{\sqrt{\chi^2(\psi)}}, \tag{119}$$

$$k_1 \geq \sqrt{\phi_1 2} \frac{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}} \psi(x_1,0)\left(\mathbb{D}_Z(x_1,0)-\mathbb{D}_Y(x_1,0)\right)}{\sqrt{\chi^2(\psi)}}, \tag{120}$$

$$k_2 \geq \frac{\phi_1\sqrt{2}}{\sqrt{\phi_2}} \frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}} \psi(0,x_2)\left(\mathbb{D}_Z(0,x_2)-\mathbb{D}_Y(0,x_2)\right)}{\sqrt{\chi^2(\psi)}}, \tag{121}$$

$$\begin{aligned} k_1+k_2 \geq{}& \sqrt{\phi_1 2} \frac{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}} \psi(x_1,0)\left(\mathbb{D}_Z(x_1,0)-\mathbb{D}_Y(x_1,0)\right)}{\sqrt{\chi^2(\psi)}} \\ &+ \frac{\phi_1\sqrt{2}}{\sqrt{\phi_2}} \frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}} \psi(0,x_2)\left(\mathbb{D}_Z(0,x_2)-\mathbb{D}_Y(0,x_2)\right)}{\sqrt{\chi^2(\psi)}}. \end{aligned} \tag{122}$$

Clearly Condition (122) is redundant in view of Conditions (120) and (121) and can be omitted. Setting $\beta_1 = \sqrt{\phi_1}$ and $\beta_2 = \frac{\phi_2}{\sqrt{\phi_1}}$ and remarking that we can relax the equalities on the message rates into $\leq$-inequalities (we can always decide to send dummy information bits) proves the desired achievability result.

### B. Converse:

Consider a sequence of length-$n$ codes with vanishing probability and vanishing covertness constraints $\delta_n \to 0$. Consider now a fixed blocklength $n$, and let $X_1^n, X_2^n$ be the random inputs generated under the chosen codes and $Y^n$ as well as $Z^n$ the corresponding outputs at the legitimate Rx and the warden.

Define

$$\alpha_{n,j,i}(x) \triangleq \Pr[X_{j,i}=x], \quad i\in\{1,\dots,n\},\ \forall j\in\{1,2\}, \tag{123}$$

and the derived nonnegative quantities

$$\alpha_n \triangleq \frac{1}{n}\sum_{i=1}^n \left( \sum_{x_1\in\mathcal{X}_1\setminus\{0\}} \alpha_{n,1,i}(x_1) + \sum_{x_2\in\mathcal{X}_2\setminus\{0\}} \alpha_{n,2,i}(x_2) \right), \tag{124}$$

and

$$\psi_n(x_1,0) = \frac{\frac{1}{n}\sum_{i=1}^n \alpha_{n,1,i}(x_1)}{\alpha_n}, \tag{125}$$

$$\psi_n(0,x_2) = \frac{\frac{1}{n}\sum_{i=1}^n \alpha_{n,2,i}(x_2)}{\alpha_n}. \tag{126}$$

The converse proof relies on the following Lemma which is an extension of [13, Lemma 1].

*Lemma 2:* Let $(X_1,X_2,Y)\sim P_{X_{1,n}}P_{X_{2,n}}\Gamma_{Y|X_1X_2}$ for some $P_{X_{2,n}}$ and some $P_{X_{1,n}}$ with $P_{X_{1,n}}(x_1)\to 0$ and $P_{X_{2,n}}(x_2)\to 0$ as $n\to\infty$ for all $x_1\in\mathcal{X}_1\setminus\{0\}$ and $x_2\in\mathcal{X}_2\setminus\{0\}$.

For any $n\in\mathbb{N}^*$ we have:

$$I(X_1;Y|X_2) \leq (1+o(1)) \sum_{x_1\in\mathcal{X}_1\setminus\{0\}} P_{X_{1,n}}(x_1)\cdot\mathbb{D}_Y(x_1,0) \tag{127}$$

$$I(X_2;Y|X_1) \leq (1+o(1)) \sum_{x_2\in\mathcal{X}_2\setminus\{0\}} P_{X_{2,n}}(x_2)\cdot\mathbb{D}_Y(0,x_2). \tag{128}$$

*1) Lower bound on $\delta_n$:* For a specific code $\mathcal{C}$, let $\widehat{Q}^n_{\mathcal{C}}(z^n)$ denote the warden's corresponding output distribution in (4). Then, by the convexity of the KL divergence:

$$\begin{aligned}
\delta_n &\geq \sum_{z^n} \widehat{Q}^n_{\mathcal{C}}(z^n) \log\left(\frac{\widehat{Q}^n_{\mathcal{C}}(z^n)}{\Gamma^{\otimes n}_{Z|X_1X_2}(z^n|0^n,0^n)}\right) && (129)\\
&\geq \sum_{i=1}^n \sum_{z_i} \widehat{Q}^{(i)}_{\mathcal{C}}(z_i) \log\left(\frac{\widehat{Q}^{(i)}_{\mathcal{C}}(z_i)}{\Gamma_{Z|X_1X_2}(z_i|0,0)}\right) && (130)\\
&= \sum_{i=1}^n \mathbb{D}\left(\widehat{Q}^{(i)}_{\mathcal{C}} \| \Gamma_{Z|X_1X_2}(\cdot|0,0)\right) && (131)\\
&\geq \sum_{i=1}^n \mathbb{D}\Bigg( \sum_{\substack{x_1\in\mathcal{X}_1\\ x_2\in\mathcal{X}_2}} \alpha_{n,1,i}(x_1)\alpha_{n,2,i}(x_2)\Gamma_{Z|X_1X_2}(\cdot|x_1,x_2) \Bigg\| \\
&\qquad\qquad \Gamma_{Z|X_1X_2}(\cdot|0,0)\Bigg) && (132)\\
&\geq n\mathbb{D}(\bar{Q}\|\Gamma_{Z|X_1X_2}(\cdot|0,0)) && (133)\\
&= n\sum_{z\in\mathcal{Z}} \Gamma_{Z|X_1X_2}(z|0,0)\left(1+\frac{\alpha_n\zeta_n(z)}{\Gamma_{Z|X_1X_2}(z|0,0)}\right) \\
&\qquad\qquad \cdot\log\left(1+\frac{\alpha_n\zeta_n(z)}{\Gamma_{Z|X_1X_2}(z|0,0)}\right) && (134)\\
&\geq n\frac{\alpha_n^2}{2}\sum_{z\in\mathcal{Z}} \frac{\zeta_n^2(z)}{\Gamma_{Z|X_1X_2}(z|0,0)} && (135)
\end{aligned}$$

where we defined $\widehat{Q}^{(i)}_{\mathcal{C}}(z_i)$ as the probability of the event $Z_i = z_i$ and

$$\zeta_n(z) \triangleq \frac{\bar{Q}(z)-\Gamma_{Z|X_1X_2}(z|0,0)}{\alpha_n} \tag{136}$$

and

$$\bar{Q}(z) \triangleq \frac{1}{n}\sum_{i=1}^n \sum_{\substack{x_1\in\mathcal{X}_1\\ x_2\in\mathcal{X}_2}} \alpha_{n,1,i}(x_1)\alpha_{n,2,i}(x_2)\Gamma_{Z|X_1X_2}(\cdot|x_1,x_2). \tag{137}$$

Since $\lim_{n\to\infty}\delta_n = 0$, by (132) we can conclude that

$$\lim_{n\to\infty} \alpha_{n,j,i}(x_j) = 0, \quad \forall i\in\{1,\dots,n\},\ j\in\{1,2\}, \quad \forall x_j\in\mathcal{X}_j\setminus\{0\}, \tag{138}$$

and hence

$$\lim_{n\to\infty} \alpha_{n,j,i}(0) = 1, \quad \forall i\in\{1,\dots,n\},\ j\in\{1,2\}. \tag{139}$$

We now derive the asymptotics of $\zeta_n$ which are going to serve later when we take limits.

*Asymptotics of $\zeta_n$:* First notice the sequence of equalities in (140) on top of the next page where we used the limiting behaviours (138) and (139) and the equalities $\alpha_{n,j,i}(0) = 1-\sum_{x_j\in\mathcal{X}_j\setminus\{0\}}\alpha_{n,j,i}(x_j)$, for $j\in\{1,2\}$.

Consider now a subsequence of blocklengths $\{n_\ell\}_{\ell=1}^\infty$ for which all sequences $\{\psi_n(x_1,0)\}$ and $\{\psi_n(0,x_2)\}$ converge for

$$\zeta_n(z) = \frac{\bar{Q}(z) - \Gamma_{Z|X_1X_2}(z|0,0)}{\alpha_n} \tag{140a}$$

$$= \sum_{(x_1,x_2)} \frac{\frac{1}{n}\sum_{i=1}^n \alpha_{n,1,i}(x_1)\alpha_{n,2,i}(x_2)}{\alpha_n} \Gamma_{Z|X_1X_2}(z|x_1,x_2) - \frac{1}{\alpha_n}\Gamma_{Z|X_1X_2}(z|0,0) \tag{140b}$$

$$= (1+o(1)) \cdot \Bigg[ \sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} \psi_n(x_1,0) \cdot \big(\Gamma_{Z|X_1X_2}(z|x_1,0) - \Gamma_{Z|X_1X_2}(z|0,0)\big) + \sum_{x_2 \in \mathcal{X}_2 \setminus \{0\}} \psi_n(0,x_2) \cdot \big(\Gamma_{Z|X_1X_2}(z|0,x_2) - \Gamma_{Z|X_1X_2}(z|0,0)\big) \Bigg], \tag{140c}$$

all inputs $x_1 \in \mathcal{X}_1$ and $x_2 \in \mathcal{X}_2$ and denote the respective convergence points by

$$\psi(x_1,0) \triangleq \lim_{\ell \to \infty} \psi_{n_\ell}(x_1,0) \tag{141a}$$

$$\psi(0,x_2) \triangleq \lim_{\ell \to \infty} \psi_{n_\ell}(0,x_2). \tag{141b}$$

Notice that $\psi(\cdot,\cdot)$ forms a pmf over the alphabet $\tilde{\mathcal{X}}$ and that we can write:

$$\zeta(z) := \lim_{\ell \to \infty} \zeta_{n_\ell}(z) \tag{142}$$

$$= \sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} \psi(x_1,0)\big(\Gamma_{Z|X_1X_2}(z|x_1,0) - \Gamma_{Z|X_1X_2}(z|0,0)\big) + \sum_{x_2 \in \mathcal{X}_2 \setminus \{0\}} \psi(0,x_2)\big(\Gamma_{Z|X_1X_2}(z|0,x_2) - \Gamma_{Z|X_1X_2}(z|0,0)\big). \tag{143}$$

*2) Upper bound on $m_1$ and $m_2$:* For a given blocklength $n$:

$$P_e \triangleq \Pr\left[\hat{W}_1 \neq W_1 \text{ or } \hat{W}_2 \neq W_2\right] \tag{144}$$

Since the message $W_1$ is uniform over $\{1,\dots,2^{m_1}\}$ and independent of the local randomness $C_1, C_2$, we have

$$m_1 = \mathbb{H}(W_1) \tag{145}$$

$$= \mathbb{H}(W_1|W_2,S_1,S_2) \tag{146}$$

$$= I(W_1;Y^n|W_2,S_1,S_2) + \mathbb{H}(W_1|Y^n,W_2,S_1,S_2,C_1,C_2) \tag{147}$$

$$\leq I(W_1;Y^n|W_2,S_1,S_2,C_1,C_2,X_2^n) + \mathbb{H}_b(P_e) + P_e \cdot m_1 \tag{148}$$

$$= \frac{1}{1-P_e}(I(W_1;Y^n|W_2,S_1,S_2,C_1,C_2,X_2^n) + \mathbb{H}_b(P_e)) \tag{149}$$

$$\leq \frac{1}{1-P_e}\left(\sum_{i=1}^n \mathbb{H}(Y_i|X_{2i}) - \mathbb{H}(Y_i|X_{1i},X_{2i}) + \mathbb{H}_b(P_e)\right) \tag{150}$$

$$= \frac{1}{1-P_e}\left(n\sum_{i=1}^n \frac{1}{n} I(X_{1i};Y_i|X_{2i}) + \mathbb{H}_b(P_e)\right). \tag{151}$$

Define $T$ as the time random variable which is uniform over $\{1,\dots,n\}$ and independent of all other random variables. By upper-bounding the binary entropy, we can rewrite (151) as:

$$m_1 \leq \frac{1}{1-P_e}\left(nI(X_{1,T};Y|X_{2,T},T) + 1\right) \tag{152}$$

$$\leq \frac{n}{1-P_e} \cdot (1+o(1)) \cdot \left(\mathbb{E}_{P_T}\left[\sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} P_{X_{1,T}|T}(x_1|T) \cdot \mathbb{D}_Y(x_1,0)\right] + \frac{1}{n}\right) \tag{153}$$

$$= \frac{n}{1-P_e} \cdot (1+o(1)) \cdot \left(\sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} P_{X_{1,T}}(x_1) \cdot \mathbb{D}_Y(x_1,0) + \frac{1}{n}\right), \tag{154}$$

where the inequality is obtained by applying Lemma 2 for each realization of $T$ and by noticing the joint pmf

$$P_{X_{1,T}X_{2,T}Y_TT}(x_1,x_2,y,t) = P_T(t)P_{X_{1,T}|T}(x_1|t)P_{X_{2,T}|T}(x_2|t)\Gamma_{Y|X_1X_2}(y|x_1,x_2) \tag{155}$$

for $P_T$ uniform over $\{1,\dots,n\}$.

By definitions (141), we obtain for the blocklengths $\{n_\ell\}$:

$$m_1 \leq \frac{n_\ell \alpha_{n_\ell}}{1-P_e}\left(\sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} \psi(x_1,0) \cdot \mathbb{D}_Y(x_1,0)\right) \cdot (1+o(1)). \tag{156}$$

Similarly, for $m_2$ and blocklengths $\{n_\ell\}$, we have:

$$m_2 \leq \frac{n_\ell \alpha_{n_\ell}}{1-P_e}\left(\sum_{x_2 \in \mathcal{X}_2 \setminus \{0\}} \psi(0,x_2) \cdot \mathbb{D}_Y(0,x_2)\right) \cdot (1+o(1)). \tag{157}$$

*3) Upper bound on $r_j$, for $j \in \{1,2\}$:* Combining (156) with (135) and (143), we obtain:

$$\lim_{\ell \to \infty} \frac{m_1}{\sqrt{n_\ell \delta_{n_\ell}}} \leq \lim_{\ell \to \infty} \frac{n_\ell \alpha_{n_\ell}}{1-P_e} \frac{\sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} \psi(x_1,0) \cdot \mathbb{D}_Y(x_1,0)}{\sqrt{n_\ell^2 \cdot \frac{\alpha_{n_\ell}^2}{2} \sum_{z \in \mathcal{Z}} \frac{\zeta_{n_\ell}^2(z)}{\Gamma_{Z|X_1X_2}(z|0,0)}}} \cdot (1+o(1)) \tag{158}$$

$$= \sqrt{2}\frac{\sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} \psi(x_1,0) \cdot \mathbb{D}_Y(x_1,0)}{\sqrt{\sum_{z \in \mathcal{Z}} \frac{\zeta^2(z)}{\Gamma_{Z|X_1X_2}(z|0,0)}}} \tag{159}$$

$$= \sqrt{2}\frac{\sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} \psi(x_1,0) \cdot \mathbb{D}_Y(x_1,0)}{\sqrt{\chi^2(\psi)}}, \tag{160}$$

and similarly

$$\lim_{\ell\to\infty}\frac{m_2}{\sqrt{n_\ell\delta_{n_\ell}}}\leq\sqrt{2}\frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi(0,x_2)\cdot\mathbb{D}_Y(0,x_2)}{\sqrt{\chi^2(\psi)}}. \tag{161}$$

*4) Upper bound on $\delta_n$:* Let $\beta_1$ and $\beta_2$ be the two numbers in $[0,1]$ that satisfy

$$\lim_{\ell\to\infty}\frac{m_1}{\sqrt{n_\ell\delta_{n_\ell}}}=\beta_1\sqrt{2}\frac{\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi(x_1,0)\cdot\mathbb{D}_Y(x_1,0)}{\sqrt{\chi^2(\psi)}} \tag{162a}$$

$$\lim_{\ell\to\infty}\frac{m_2}{\sqrt{n_\ell\delta_{n_\ell}}}=\beta_2\sqrt{2}\frac{\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi(0,x_2)\cdot\mathbb{D}_Y(0,x_2)}{\sqrt{\chi^2(\psi)}}. \tag{162b}$$

Assume for the moment that $\beta_1,\beta_2$ are strictly larger than 0, and thus we can divide by them. Then, (162) combined with (156) and (157) implies for all blocklengths $n_\ell$:

$$\sqrt{n_\ell\delta_{n_\ell}}\leq\frac{n_\ell\alpha_{n_\ell}}{\beta_j\sqrt{2}(1-P_e)}\sqrt{\chi^2(\psi)}(1+o(1)),\quad j\in\{1,2\}. \tag{163}$$

*5) Lower bound on $m_j+p_j$, for $j\in\{1,2\}$:* Using standard steps, one can show that:

$$\begin{aligned}m_1+p_1&=\mathbb{H}(W_1,S_1|C_1,C_2) &(164)\\ &\geq I(W_1,S_1;Z^n|C_1,C_2) &(165)\\ &\geq I(X_1^n;Z^n|C_1,C_2) &(166)\\ &\geq I(X_1^n,X_2^n;Z^n|C_1,C_2)-I(X_2^n;Z^n|X_1^n), &(167)\end{aligned}$$

and similarly

$$m_2+p_2\geq I(X_1^n,X_2^n;Z^n|C_1,C_2)-I(X_1^n;Z^n|X_2^n). \tag{168}$$

We next define for each pair $(c_1,c_2)\in\mathcal{G}_1\times\mathcal{G}_2$:

$$\begin{aligned}&\widehat{Q}^n_{\mathcal{C},(c_1,c_2)}(z^n)\\ &\triangleq\frac{\sum_{(w_1,s_1)}\sum_{(w_2,s_2)}\Gamma^{\otimes n}_{Z|X_1X_2}(z^n|x_1^n(w_1,s_1,c_1),x_2^n(w_2,s_2,c_2))}{2^{m_1+m_2+p_1+p_2}}\end{aligned} \tag{169}$$

and the divergence

$$\delta_{n,(c_1,c_2)}\triangleq\mathbb{D}\left(\widehat{Q}^n_{\mathcal{C},(c_1,c_2)}\,\|\,\Gamma^{\otimes n}_{Z|X_1X_2}(\cdot|0^n,0^n)\right). \tag{170}$$

With these definitions, we can write:

$$\begin{aligned}&I(X_1^n,X_2^n;Z^n|C_1,C_2) &(171)\\ &=\mathbb{E}\left[\mathbb{D}\left(\Gamma^{\otimes n}_{Z|X_1X_2}(\cdot|X_1^n,X_2^n)\|\widehat{Q}^n_{\mathcal{C},(C_1,C_2)}\right)\right] &(172)\\ &=\mathbb{E}\Bigg[\sum_{z^n}\Gamma^{\otimes n}_{Z|X_1X_2}(z^n|X_1^n,X_2^n)\\ &\qquad\cdot\log\left(\frac{\Gamma^{\otimes n}_{Z|X_1X_2}(z^n|X_1^n,X_2^n)}{\Gamma^{\otimes n}_{Z|X_1X_2}(z^n|0^n,0^n)}\right)\Bigg]\\ &\quad-\mathbb{E}\left[\delta_{n,(C_1,C_2)}\right] &(173)\\ &\overset{(a)}{\geq}\sum_{i=1}^n\mathbb{E}\Bigg[\sum_{z_i}\Gamma_{Z|X_1X_2}(z_i|X_{1,i},X_{2,i})\\ &\qquad\cdot\log\left(\frac{\Gamma_{Z|X_1X_2}(z_i|X_{1i},X_{2i})}{\Gamma_{Z|X_1X_2}(z_i|0,0)}\right)\Bigg]-\delta_n &(174)\\ &\overset{(b)}{=}\sum_{i=1}^n\Bigg(\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\alpha_{n,1,i}(x_1)\mathbb{D}_Z(x_1,0)\\ &\qquad+\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\alpha_{n,2,i}(x_2)\mathbb{D}_Z(0,x_2)\Bigg)\cdot(1+o(1))\\ &\quad-\delta_n &(175)\\ &=n\alpha_n\Bigg(\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi_n(x_1,0)\mathbb{D}_Z(x_1,0)\\ &\qquad+\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi_n(0,x_2)\mathbb{D}_Z(0,x_2)]\Bigg)(1+o(1))-\delta_n &(176)\end{aligned}$$

where above sequence of (in)equalities are justified as follows:

- $(a)$ holds by convexity of the divergence;
- $(b)$ by writing out the expectations over the independent random variables $X_{1,i}$ and $X_{2,i}$; by noting that for $X_{1,i}=X_{2,i}=0$ the term in the expectation evaluates to 0; and by recalling that all $\alpha_{n,1,i},\alpha_{n,2,i}\to 0$ as $n\to\infty$.

For the second term in (167), we have

$$\begin{aligned}&I(X_1^n;Z^n|X_2^n)\\ &=\mathbb{H}(Z^n|X_2^n)-\mathbb{H}(Z^n|X_1^n,X_2^n) &(177)\\ &=\sum_{i=1}^n\mathbb{H}(Z_i|Z^{i-1},X_2^n)-\mathbb{H}(Z_i|Z^{i-1},X_1^n,X_2^n) &(178)\\ &=\sum_{i=1}^n\mathbb{H}(Z_i|X_2^n)-\mathbb{H}(Z_i|X_{1,i},X_{2,i},X_{3,i}) &(179)\\ &\leq\sum_{i=1}^n\mathbb{H}(Z_i|X_{2i})-\mathbb{H}(Z_i|X_{1i},X_{2i}) &(180)\\ &=n\sum_{i=1}^n\frac{1}{n}I(X_{1i};Z_i|X_{2i}) &(181)\\ &=nI(X_{1,T};Z_T|X_{2,T},T) &(182)\\ &\leq n\alpha_n\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi_n(x_1,0)\mathbb{D}_Z(x_1,0)\\ &\qquad\cdot(1+o(1)), &(183)\end{aligned}$$

and

$$I(X_2^n;Z^n|X_1^n)\leq n\alpha_n\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi_n(0,x_2)\mathbb{D}_Z(0,x_2)\cdot(1+o(1)). \tag{184}$$

By (176), (183), (184), (167), and (168), we conclude

$$m_1+p_1\geq n\alpha_n\sum_{x_1\in\mathcal{X}_1\setminus\{0\}}\psi_n(x_1,0)\mathbb{D}_Z(x_1,0)(1+o(1))-\delta_n \tag{185}$$

$$m_2+p_2\geq n\alpha_n\sum_{x_2\in\mathcal{X}_2\setminus\{0\}}\psi_n(0,x_2)\mathbb{D}_Z(0,x_2)(1+o(1))-\delta_n\quad. \tag{186}$$

*6) Lower bound on $r_j + k_j$, for $j \in \{1,2\}$:* Restricting to the subsequence of blocklengths $\{n_\ell\}$, by the definition of $\psi(x_1, 0)$ and $\psi(0, x_2)$ and because $\delta_n$ is negligible compared to $n\alpha_n$, we obtain from (185), (186), and (163):

$$\lim_{\ell\to\infty} \frac{m_1 + p_1}{\sqrt{n_\ell \delta_{n_\ell}}} \geq \beta_1 \sqrt{2} \frac{\sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} \psi(x_1, 0) \mathbb{D}_Z(x_1, 0)}{\sqrt{\chi^2(\psi)}} \quad (187)$$

$$\lim_{\ell\to\infty} \frac{m_2 + p_2}{\sqrt{n_\ell \delta_{n_\ell}}} \geq \beta_2 \sqrt{2} \frac{\sum_{x_2 \in \mathcal{X}_2 \setminus \{0\}} \psi(0, x_2) \mathbb{D}_Z(0, x_2)}{\sqrt{\chi^2(\psi)}}. \quad (188)$$

*7) Concluding the Proof:* Notice that so far we have assumed that $\beta_1, \beta_2 > 0$. Obtaining Bounds (187) and (188) for $\beta_1$ or $\beta_2$ zero is trivial by the nonnegativity of the quantities. The proof is then obtained in a rather straightforward way by collecting all established results.

■

### C. Proof of Theorem 5

In Proposition 10, for any given $\psi(\cdot)$ and $(k_1, k_2)$, we can optimize over all permissible choices of $\beta_1, \beta_2 \in [0, 1]$ satisfying (76) and (77) so as to maximize the rate constraints (74) and (75). For a given $\Delta_\psi > 0$ and

$$k_1 \in \left[0,\ \sqrt{2} \frac{\sum_{x_1 \in \mathcal{X}_1 \setminus \{0\}} \psi(x_1, 0) \cdot (\mathbb{D}_Z(x_1, 0) - \mathbb{D}_Y(x_1, 0))}{\sqrt{\chi^2(\psi)}}\right] \quad (189)$$

$$k_2 \in \left[0,\ \sqrt{2} \frac{\sum_{x_2 \in \mathcal{X}_2 \setminus \{0\}} \psi(0, x_2) \cdot (\mathbb{D}_Z(0, x_2) - \mathbb{D}_Y(0, x_2))}{\sqrt{\chi^2(\psi)}}\right] \quad (190)$$

the optimal choice of $\beta_1$ and $\beta_2$ is to turn Inequalities (76) and (77) into equalities. By our assumptions (189) and (190), the corresponding $\beta_1, \beta_2$ values lie in $[0, 1]$. In contrast, for key-rates $k_1$ and $k_2$ outside above ranges (189) and (190) or when $\Delta_\psi \leq 0$, the optimal choice is $\beta_1 = 1$ and $\beta_2 = 1$.

Plugging these optimal and permissible choices of $\beta_1, \beta_2$ into the rate expressions (74) and (75) of Proposition 10, we obtain for each pmf $\psi$ that the pair $(r_1, r_2)$ is achievable if, and only if, it lies in the region $\mathcal{R}^\star_\psi(k_1, k_2)$. This concludes the proof of Theorem 5.

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